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Systems and methods for offering a plurality of electronic table games of chance on an electronic gaming machine including side bets to increase hold

Abstract

A system and method for operating a plurality of games of chance. The gaming system includes a primary game display area and a secondary game display area. The plurality of games includes at least a plurality of electronic table games including two or more electronic table games from a group of table games including, craps, baccarat, roulette, blackjack, three-card poker, pai gow poker, Caribbean stud and big 6 wheel. The processing of the two or more electronic table games includes a separate and discrete game state engine for each of the two or more electronic table games. High hold percentage side bets are offered to increase the hold percentage of at least one table game and the overall hold percentage of the gaming system.

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Background/Summary

CROSS-REFERENCES (1) This application is a continuation of, and claims priority to, U.S. patent application Ser. No. 16/934,956 filed on Jul. 21, 2020 and which is incorporated herein for all purposes.

FIELD OF THE INVENTION
(1) The embodiments of the present invention relate to systems and methods for offering a plurality of table games of chance including side wagers on an electronic gaming machine while offering configurability options to more closely emulate actual table games utilizing multiple discrete game engines with separate game states and code paths including code loops.
BACKGROUND
(2) Casinos derive much of their gaming revenue from electronic gaming machines (“EGMs”) such as slot machines. Unfortunately, even with the introduction of new technology (e.g., curved displays), slot machines and the like can become stale after even short game play sessions. Moreover, younger players do not tend to play traditional slot machines because they are not exciting or attractive to play. Often such players find casino table games more exciting than traditional slot machines. Therefore, as the player demographic continues to trend younger, new

and exciting electronic table games of chance are needed.
(3) Many conventional casino table games exist with many being staples of most casinos. These primary table games include craps, roulette, blackjack, and baccarat with many other proprietary or secondary table games such as Pai Gow Poker, Three Card Poker, Caribbean Stud, etc., to offer players a wide range of game choices. While casino table games include a house advantage, the advantage is generally far less than the hold percentages of slot machines. For instance, a knowledgeable craps player can reduce the house advantage to less than 1% by utilizing the proper strategy of pass or come line bets with odds. Similarly, experienced players of blackjack, also known as 21, can often reduce the house percentage to under 1% using skilled play. Other table games such as roulette and baccarat have fixed house percentages of 5.26% for double zero roulette and 1.06% and 1.24% for baccarat, depending on wagering on the player or banker. These house percentages are far less than the average RTP which most EGMs are set at, usually ranging between 94% and 86% meaning the corresponding house advantage is 6% to 14%. Some table games offer a variety of bets which help offset these low house advantages. For instance, many craps wagers outside of pass and come line wagers, can have house advantages ranging between just over 1% to over 16% which can offset the low hold percentages of pass and come line bets. To offset the low house advantages in other table games such as blackjack and baccarat, side bets have been developed with hold percentages approaching 15% which when taken with standard wagers can significantly increase the house advantage.
(4) The need for casinos to set RTP of EGMs at a low rate is dependent on total coin-in per day (total amount wagered) and cost of the EGMs which can approach or even exceed $20,000 per machine. Accordingly, if table games are produced in an electronic form, it is advantageous to increase hold percentages so they approach the RTP of traditional EGMs. Although some table games have been converted to EGMs, such as roulette and blackjack, it is often very difficult to have comparable win per unit (WPU) with traditional EGMs due to limited play and low hold percentages.
(5) Accordingly, the new system and method detailed herein involve producing electronic versions of table games that reside on a common multi-game platform to achieve the volume of play necessary to warrant the cost of the electronic table game (ETG) and to produce comparable RTP percentages. Creative side bets with higher hold percentages assist in this effort. Moreover, most casino pits (the area of a casino which typically houses and operates table games) may have restricted play due to health concerns such as the COVID-19 pandemic of 2020. This restricted play, known as social distancing, may eliminate half or more of the gaming positions in the casino pit thereby often giving table games players little or no options if their desired table game is full. The embodiments of the present invention described herein provide options in such environments while increasing the hold and offering a multitude of table game options. To further emulate the table games in the pit, the embodiments of the present invention also provide configuration options allowing casino operators to configure the ETG similar to the wager limits and rules of the table games offered in their pit.

SUMMARY

(6) The embodiments of the present invention relate to systems and methods for offering a plurality of table games of chance including side wagers on an electronic gaming machine while offering configurability options to more closely emulate table games in a casino pit.
(7) In one embodiment, each table game has a dedicated game engine for controlling play of the selected table game.
(8) Other variations, embodiments and features of the present invention will become evident from the following detailed description, drawings and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** illustrates an exemplary gaming machine of the type that may be used to facilitate the embodiments of the present invention;

(2) FIG. **2** illustrates an exemplary kiosk of the type that may be used to facilitate the embodiments of the present invention;

(3) FIG. **3** illustrates a block diagram of a multiple casino property system of the type that may be used to facilitate the embodiments of the present invention;

(4) FIG. **4** illustrates a block diagram of a wireless network system of the type that may be used to facilitate the embodiments of the present invention;

(5) FIG. **5** illustrates a diagram of exemplary components of a computing device of the type that may be used to facilitate the embodiments of the present invention;

(6) FIG. **6** illustrates a diagram of exemplary gaming device hardware of the type that may be used to facilitate the embodiments of the present invention;

(7) FIG. **7** illustrates a diagram of gaming device program modules of the type that may be used to facilitate the embodiments of the present invention;

(8) FIG. **8** illustrates a user interface presented to a player wherein the player selects one of a plurality of table games according to the embodiments of the present invention;

(9) FIG. **9**A illustrates a user interface of a LCD button deck of the electronic table game of craps according to the embodiments of the present invention;

(10) FIG. **9**B illustrates the top display of an EGM relating to the electronic table game of craps according to the embodiments of the present invention;

(11) FIG. **9**C illustrates the bottom display of an EGM relating to the electronic table game of craps according to the embodiments of the present invention;

(12) FIG. **10**A illustrates a user interface of a LCD button deck of the electronic table game of baccarat according to the embodiments of the present invention;

(13) FIG. **10**B illustrates the top display of an EGM relating to the electronic table game of baccarat according to the embodiments of the present invention;

(14) FIG. **10**C illustrates the bottom display of an EGM relating to the electronic table game of baccarat according to the embodiments of the present invention;

(15) FIG. **11**A illustrates a user interface of a LCD button deck of the electronic table game of roulette according to the embodiments of the present invention;

(16) FIG. **11**B illustrates the top display of an EGM relating to the electronic table game of roulette according to the embodiments of the present invention;

(17) FIG. **11**C illustrates the bottom display of an EGM relating to the electronic table game of roulette according to the embodiments of the present invention;

(18) FIG. **12**A illustrates a user interface of a LCD button deck of the electronic table game of blackjack according to the embodiments of the present invention;

(19) FIG. **12**B illustrates the top display of an EGM relating to the electronic table game of blackjack according to the embodiments of the present invention;

(20) FIG. **12**C illustrates the bottom display of an EGM relating to the electronic table game of blackjack according to the embodiments of the present invention;

(21) FIG. **13** illustrates a block diagram depicting discrete game engines for a plurality of electronic table games according to the embodiments of the present invention;

(22) FIG. **14** illustrates an expanded block diagram of the craps game engine of FIG. **13** for electronic table games of the present invention;

(23) FIG. **15** illustrates an expanded block diagram of the baccarat game engine of FIG. **13** for electronic table games of the present invention;

(24) FIG. **16** illustrates an expanded block diagram of the roulette game engine of FIG. **13** for electronic table games of the present invention;

(25) FIG. **17** illustrates an expanded block diagram of the blackjack game engine of FIG. **13** for electronic table games of the present invention;
(26) FIG. **18** illustrates a combined single display of the embodiment illustrated in FIG. **9** relating to the table game of craps according to the embodiments of the present invention;
(27) FIG. **19** illustrates a combined single display of the embodiment illustrated in FIG. **10** relating to the table game of baccarat according to the embodiments of the present invention;
(28) FIG. **20** illustrates a combined single display of the embodiment illustrated in FIG. **11** relating to the table game of roulette according to the embodiments of the present invention; and
(29) FIG. **21** illustrates a combined single display of the embodiment illustrated in FIG. **12** relating to the table game of blackjack according to the embodiments of the present invention.

DETAILED DESCRIPTION

(30) For the purposes of promoting an understanding of the principles in accordance with the embodiments of the present invention, reference will now be made to the embodiments illustrated in the drawings and specific language will be used to describe the same. It will nevertheless be understood that no limitation of the scope of the invention is thereby intended. Any alterations and further modifications of the inventive feature illustrated herein, and any additional applications of the principles of the invention as illustrated herein, which would normally occur to one skilled in the relevant art and having possession of this disclosure, are to be considered within the scope of the invention claimed.
(31) Those skilled in the art will recognize that the embodiments of the present invention involve both hardware and software elements, which portions are described below in such detail required to construct and operate a game method and system according to the embodiments of the present invention.
(32) As will be appreciated by one skilled in the art, aspects of the present invention may be embodied as a system, method and/or computer program product. Accordingly, aspects of the present invention may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.), or an embodiment combining software and hardware. Furthermore, aspects of the present invention may take the form of a computer program product embodied in one or more computer readable medium(s) having computer readable program code embodied thereon.
(33) Any combination of one or more computer readable medium(s) may be utilized. The computer readable medium may be a computer readable signal medium or a computer readable storage medium. A computer readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples (a non-exhaustive list) of the computer readable storage medium would include the following: an electrical connection having one or more wires, a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), an optical fiber, a portable compact disc read-only memory (CD-ROM), and optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer readable storage medium may be any tangible medium that can contain or store a program for use by or in connection with an instruction execution system, apparatus, or device.
(34) A computer readable signal medium may include a propagated data signal with computer readable program code embodied thereon, for example, in baseband or as part of a carrier wave. Such a propagated signal may take any variety of forms, including, but not limited to, electromagnetic, optical, or any suitable combination thereof. A computer readable signal medium may be any computer readable medium that is not a computer readable storage medium and that can communicate, propagate, or transport a program for use by or in conjunction with an instruction execution system, apparatus, or device.

(35) Program code embodied on a computer readable medium may be transmitted using any appropriate medium, including but not limited to wired, wireless, wireline, optical fiber cable, RF, Bluetooth and the like, or any suitable combination of the foregoing.
(36) Computer program code for carrying out operations for aspects of the present invention may be written in any combination of one or more programming languages, including an object-oriented programming language such as Java, Smalltalk, C++ or the like or conventional procedural programming languages, such as the “C” programming language, AJAX, PHP, HTML, XHTML, Ruby, CSS or similar programming languages. The programming code may be configured in an application, an operating system, as part of a system firmware, or any suitable combination thereof. The programming code may execute entirely on the user's computer, partly on the user's computer, as a standalone software package, partly on the user's computer and partly on a remote computer or entirely on a remote computer or server as in a client/server relationship sometimes known as cloud computing. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider).
(37) Aspects of the present invention are described below with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems) and computer program products according to embodiments of the invention. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general-purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flowchart and/or block diagram.
(38) These computer program instructions may also be stored in a computer readable medium that can direct a computer, other programmable data processing apparatus, or other devices to function in a particular manner, such that the instructions stored in the computer readable medium produce an article of manufacture including instructions which implement the function/act specified in the flowchart and/or block diagram.
(39) The computer program instructions may also be loaded onto a computer, other programmable data processing apparatus, or other devices to cause a series of operational steps to be performed on the computer, other programmable apparatus or other devices to produce a computer-implemented process such that the instructions which execute on the computer or other programmable apparatus provide processes for implementing the functions/acts specified in the flowchart and/or block diagrams. As used herein, a “gaming machine” should be understood to be any one of a general purpose computer, as for example a personal computer, laptop computer, standalone machine, a client computer configured for interaction with a server, a special purpose computer such as a server, or a smart phone, soft phone, tablet computer, personal digital assistant or any other machine adapted for executing programmable instructions in accordance with the description thereof set forth above.
(40) Those skilled in the art will recognize that certain types of EGMs, generally utilized in regulated casino environments, are still commonly referred to as “slot machines”. Although the etymology of the term “slot machine” was originally derived from a coin slot in the gaming machines at the time, coin slots have long since generally been replaced by payment input devices or bill acceptors, also known as bill validators, which only accept paper currency or ticket-in-ticket-out vouchers and/or electronic fund transfer means, such as card readers, mobile device payment means or account interfaces. As a result, the term EGM and slot machine are used interchangeably and are defined to mean something different than a laptop or desktop computer, cell phones, tablet

computer gaming devices and the like.
(41) EGMs may be classified as Class II, Class III, video lottery terminals (VLT), or the like. EGMs may present either one or a plurality of games to the player such as video reels, video poker, video keno, video bingo, and the like. In another embodiment, the gaming devices are gaming kiosks or terminals. Alternatively, the gaming devices may include remote gaming devices, for example, cellular phones, laptop or desktop computers, and/or any other suitable devices. The servers may include one or more local servers within a gaming establishment and/or one or more wide area progressive (WAP) servers connected to the local servers and/or to the gaming devices through the network.
(42) In one embodiment, each gaming device presents either one or a plurality of games of chance to a player to enable the player to select and play the games of chance. In addition, each gaming device may include a randomization device, such as a random number generator (RNG) and/or a permutation generator, that is used to play a selected game on the gaming device. The randomization device may be used to randomly determine a game outcome for the game of chance. For example, if the player selects a game of bingo to be played on a gaming device, the gaming device uses the randomization device to select a plurality of house indicia from a pool of indicia to be used during the game. In another embodiment, at least some aspects of the game are provided by one or more servers. The server or servers may include a randomization device for randomly selecting the house indicia in the bingo game or any other wagering event.
(43) In the example of a video poker game, either one or a plurality of poker games are presented to the player. After game selection and wagering, a number of playing cards, generally selected from a 52-card deck, are distributed to the player. In the case of draw poker or its many variants, the player selectively chooses to retain one or more of the original cards dealt and to discard those cards not chosen to be retained. The discarded cards are then replaced by new cards. If the player obtains a predefined winning combination of cards, the player wins an amount associated with the particular winning combination of cards.
(44) In the example of mechanical, electromechanical, or video reel machines, the games may include a number of mechanical or simulated rotating reels that are arranged in a horizontal configuration forming columns or vertical configurations forming rows. Alternatively, simulated rotating reels may be arranged in a vertical configuration forming columns or vertical configurations forming rows. One or a number of rows are presented to the player to allow for one or many different winning pay lines. Pay lines may be straight across or designed in any convenient fashion. A typical game many include five reels or columns and three or four rows or the like or a vertical configuration of five rows and three or four columns and the like.
(45) In the example of the bingo game, the house indicia are compared to a plurality of player indicia that are included within a pattern selected for one or more player cards. If at least some of the player indicia within the pattern are matched by the house indicia, the player may win a prize based on the number of house indicia that have been matched and an associated pay table.
(46) In the example of a keno game or a keno-related game of chance, the gaming device uses the randomization device to randomly select a plurality of house indicia in a similar manner as described with respect to the game of bingo. However, twenty house indicia are typically randomly selected or called from a pool of 80 house indicia, although other sizes of house indicia pools may be used. The called house indicia are compared to a plurality of player indicia to determine how many player indicia are matched by the house indicia and may be irrespective of a pattern of the player indicia. The embodiments described herein may include allowing the player to select the number of and specific player indicia to be utilized for a keno game or may include an automated or quick pick selection. For example, a player may select one player indicia or spot to play a 1 spot game, 2 player indicia or spots for a 2-spot game, 3 player indicia or spots for a 3-spot game, etc. Embodiments may also require a minimum number of player indicia or spots to match to win a game. For example, 10 player indicia or 10-spot game may require a minimum of 5 player indicia

or spots to match the randomly selected player indicia. Embodiments may also include a maximum number of player indicia or spots that are playable. For example, in an 80-number game, the maximum number of house indicia or spots selectable by the player may be confined to 20 numbers or less or a 20-number game or less. Accordingly, in an 80-number game, the minimum number of player indicia or spots may be 2 and the maximum player indicia or spots may be 20. The player may win one or more prizes based on the number of player indicia matched by the called house indicia.
(47) In the example of sports wagering, a player may be seated in a player area that may include a betting terminal which includes a monitor and input means. A player may make or place periodic wagers on a variety of sporting events.
(48) As the player plays the games, the gaming device and/or a server or another computing device tracks data representative of the gameplay of the player (referred to herein as “gameplay data”), such as a theoretical win or loss, a past history, wager amounts, a number of plays per hour, wager amounts relative to an amount of time spent playing games on the gaming device, a number of wins or losses of the player, a cumulative amount wagered by the player, an amount of money won or lost by the player, and/or any other suitable data. The gameplay data is used to determine whether the player is eligible to receive a comp. The comp may include, for example, one or more free beverages, free meals, free tickets, reduced price meals or tickets, and/or the like.
(49) In one embodiment, a comp indicator is included within, attached to, or displayed on the gaming device. The comp indicator may be energized or activated in any conventional way to indicate status including displaying on the game monitor, player tracking module or the like. The comp indicator is used to display to the player and/or to gaming establishment employees whether the player is eligible to receive the comp. If the gameplay data indicates that the player has reached a predetermined threshold of play and/or wagering activity, for example, the player is determined to be eligible to receive the comp. The comp indicator may then be activated to notify the player and/or gaming establishment employees that the player is eligible to receive the comp. The comp indicator activation may include any suitable means for displaying comp status, comp eligibility, change in comp status, incremental progress toward comps, continual progress toward comps, reduction in comp status after awarding of comps, etc., and may include any visual or sensory indicator or indication. Gaming establishment employees may then take action in response to the notification, such as by awarding the comp to the player. While the comp indicator is sometimes described as being a visual indicator, it should be recognized that the comp indicator may notify the player and/or gaming establishment employees using any suitable sensory perception, via printed comp tickets or the like.
(50) A technical effect of the systems and methods described herein includes one or more of: (a) presenting a game of chance to a player on a gaming device; (b) enabling the player to input money or credits or physical items representing money or credits for use in the game of chance using a payment input device of the gaming device; (c) enabling the player to withdraw money or credits from the gaming device using a payment output device of the gaming device; (d) providing a comp indicator attached to or integrated within the gaming device, wherein the comp indicator is configured to provide an indication if the player is determined to be eligible for a comp; (e) generating gameplay data associated with the game of chance or skill-based game of chance for the player using the gaming device; (f) receiving input from the player at the gaming device to enable the player to play the game of chance; (g) randomly determining a game outcome for the game of chance using a randomization device; (h) transmitting the gameplay data from the gaming device to a computing device; (i) determining, by the computing device, whether the player is eligible for the comp based on the gameplay data; and (j) transmitting data representative of whether the player is determined to be eligible for the comp from the computing device to the gaming device.
(51) Comp monitoring or accounting may also be monitored locally or remotely by management to ensure proper compliance. Systems and methods described herein may be self-contained within a

gaming device or may reside in a server-based system such as a slot accounting system (SAS).
(52) As used herein, a "game of chance" or "game" refers to a manual or an electronic game that is played by a player in which an outcome of the game of chance is at least partially based on chance or a random selection of game components or skill-based game components. A game may be categorized by a game variety and/or a game size, for example. It should be recognized by those of ordinary skill in the art that the term "random" is not limited to true randomness, such as truly random numbers. Rather, pseudorandom numbers and pseudorandom algorithms are included within the meaning of "random." In addition, those of ordinary skill in the art will recognize that permutation generators may additionally or alternatively be used to generate player card indicia or other game components.
(53) Gaming devices described herein may use real money for play or may utilize a credit-based system in which the credits used for the games may or may not have a cash value. Similarly, prizes for the games may be in the form of credits, cash, and/or physical prizes such as televisions, automobiles, or the like.
(54) A "local game" is a game that is played by players within a predetermined location, such as within a single gaming establishment, or players playing the game across a local area network. A "local prize" or a "local payout" (including a local progressive prize or a local progressive payout) is a prize that may be won during a local game.
(55) As used herein, the terms "connect" and "couple" are not limited to only including direct connections. Rather, unless otherwise specified, indirect connections are included within the definitions of "connect" and "couple." For example, two devices may be considered connected even if there are other devices or components connected between the two devices. Any suitable means to connect or couple devices or components together may be used.
(56) A player reward card refers to a physical or electronic card, token, or other device or data that enables a system to identify a player in connection with, among other things, a reward program or campaign. Accordingly, the player reward card may serve to identify the player and may enable gameplay, credits, funds, or other data to be associated with the player. In addition, player card tier levels may be established to denote the level of player play or relative worth to the casino operator.
(57) FIG. **1** is an illustration of an exemplary electronic gaming machine (EGM) **100** that may be used with the systems described herein. In one embodiment, EGM **100** is an electronic table games gaming device (ETG) **114**. EGM **100** may include one or more comp indicators **102**, which may be incorporated into, or implemented by, a candle device **105**, lighting element **130**, displayed on monitor **116** or **118**, displayed on the player tracking module **134**, displayed as a LED indicator on button panel **136**, or another device. One or more cameras **132** are provided with or as part of the EGM **100** to capture images of the player or other aspects of game play.
(58) The comp indicator **102** visually notifies or alerts the player or casino staff when the player is determined to be eligible to receive one or more comps from a gaming establishment, for example. The comp indicator **102** may also display or otherwise notify the player of the progress towards attaining the comp or comps. Such comps may include, for example, one or more free beverages, free meals, free rooms, free credits for one or more games of chance, free prizes, free tickets to a performance, free services (e.g., spa services), and/or a discount or reduced price for one or more of the foregoing goods or services (e.g., with respect to a market price of the goods or services). In one embodiment, comp indicator **102** may include an audio notification or other sensory notification in addition to, or in place of, the visual notification. While comp indicator **102** is described as being used with EGM **100**, it should be recognized that comp indicator may be used with any gaming device **114** and/or computing device.
(59) The EGM **100** also includes a cabinet **106** configured to support and secure the elements of the EGM. The EGM **100** includes one or more monitors/screens such as an upper screen **118** and a lower screen **116**. The screens **116** and **118** may be configured to display game content to the player or any other information regarding the game, the casino, rules, pay tables, promotions,

advertisements, or any multimedia content. Any type screen may be used, such as a flat screen or curved screen display. Those skilled in the art will recognize that games or other content may be displayed on a single display as opposed to dual displays or even split into three separate displays. Generally, when content is displayed on a single screen the display is placed in a portrait mode. Additional displays may also be utilized for a variety of purposes such as advertising toppers, wheel toppers, LCD button panels, and the like. Additional lights **130** may be incorporated into the gaming machine to providing lighting for the player or ornamentation for the EGM **100**.

(60) A scanner **108** is provided to scan tickets which have bar or box codes, or for scanning money, cards, electronic funds transfer devices, or any other media. In addition, scanner **108** may include other connectivity means such as Bluetooth communications, near field communications or similar. Similar, a card reader **112** is provided to read one or more aspects of cards, such as player tracker or rewards cards, personal identification cards, and/or credit cards. The EGM **100** may also include a printer **110**. The printer may print on any type media. Any type of content may be printed including but not limited to cash out tickets, coupons, gift certificates, comps, prizes, gaming codes, redemption codes, bar or box codes, receipts, or any other type of information. Also, part of this embodiment is a cash acceptor **104** configured to accept paper money, ticket-in-ticket-out vouchers, or any type physical item associated with the EGM **100**. A USB port **138** or other type charging or **110** port is provided for phone charging or interfacing the user's smart phone to the EGM **100**. Numerous other buttons and player interface elements are presented with the EGM **100** to accept player input. The screens **116** and **118** may be configured as touch screens.

(61) FIG. **2** is an illustration of an exemplary kiosk **200** that may be used with the systems described herein. In one embodiment, kiosk **200** is an electronic device provided for users to obtain information, conduct business, enter information, or any other use for which a computing device with communication capability is useful. The kiosk **200** may also be used for gaming for such games as keno, bingo, sports betting, etc. Unless otherwise specified, kiosk **200** shares some components and functionality with EGM **100** (shown in FIG. **1**) and similar components are labeled in FIG. **2** with the same reference numerals as used in FIG. **1**.

(62) Kiosk **200** may include one or more informational displays **202**, which may be incorporated into, or implemented by a first display **216** and/or second display **218**. Also shown in association with the kiosk **200** is a keyboard **224** which may be fixed or fold down from the front of the kiosk **200** to provide a user input device. The first display **216** and/or second display **218** may be configured as touch screens thereby allowing user input.

(63) In use, a user may use the kiosk **200** for any use now known or developed in the future. Such uses include but are not limited to, check in or check out of a hotel, spa, restaurant, gaming area, pool, or any other location or service. The kiosk **200** may also be used to sign up for an event or program, such as but not limited to a player reward program, tournament, or event. The kiosk **200** may also be used to purchase tickets, goods or services. One of ordinary skill in the art will arrive at other uses for kiosk **200**.

(64) FIG. **3** is a block diagram of a system **300** that may be used to play one or more games of chance, such as video poker, video slots, sports betting, bingo, keno or any the wagering game. The games of chance may be played by a player against other players or may be played by the player against the house.

(65) System **300** is operated using components and devices within one or more gaming establishments **302**, such as a first gaming establishment **304**, a second gaming establishment **306**, and a Nth gaming establishment **309** (not shown). It should be recognized that any suitable number of gaming establishments **302** may be provided within system **300**. Accordingly, system **300** is not limited to including two gaming establishments **302** as illustrated. In one embodiment, gaming establishments **302** are locations in which devices (e.g., gaming devices) that play or operate at least a portion of the games of chance are located. For example, gaming establishments **302** may be casinos, racetracks, bingo halls, keno parlors, or any other establishments. In another example,

gaming establishments **302** may be residences or businesses in which one or more devices are located for playing or operating the games of chance. Gaming establishments **302** may additionally or alternatively include any combination of the examples described herein.

(66) In one embodiment, gaming establishments **302** are physically remote from each other and are communicatively connected to at least one network **308**, such as a wide area network (WAN), a metropolitan area network (MAN), and/or the Internet, for example. Alternatively, the gaming establishments **302** may be separate rooms or sections of a casino or another facility that are communicatively connected by network **308**. It should be recognized that network **308** may be a wired Ethernet network, a wireless Ethernet network, a combination of wired and wireless Ethernet networks, or any other suitable wired and/or wireless network.

(67) In one embodiment, each gaming establishment **302** includes a local game server **310** (referred to herein as a “local server”) and a player reward server **312**. Local server **310** and player reward server **312** may alternatively be implemented as or within a single server. The local server **310** is coupled to a plurality of the gaming devices **314** through an internal network **316**, such as a private local area network (LAN) within the gaming establishment **302**, for example. The gaming devices **314** may be located in separate gaming establishments **302**, or within the same gaming establishment. In one embodiment, a gateway **318** is provided to enable the local server **310** of each gaming establishment **302** to securely connect to the network **308**.

(68) In one embodiment, the local server **310** is a server computer (or “server”) that monitors and controls the games played on gaming devices **314**, including local games. In one embodiment, the local games include games that are played against the house and/or that are played against other players within gaming establishment **302**.

(69) In addition, the local server **310** may administer other background tasks that enable games to be played on the gaming devices **314**. For example, the local server **310** may facilitate authenticating gaming devices **314** and the players using the gaming devices **312** and may facilitate allocating payments or credits between players and the house. The local server **310** may include cashless payment processing capabilities to enable players to receive electronic funds from a bank or another financial institution or to deposit electronic funds to the bank or financial institution. Alternatively, the payment processing capabilities may be included in a separate server or another device that is communicatively connected to the local server **310**. In addition, the local server **310** may interface with the player reward server **312** to facilitate tracking and administering player rewards. Each gaming device **314**, group of gaming devices **314**, local servers **310**, player reward servers **312**, or the like may collect and/or generate data desired for accounting purposes, such as for use in slot accounting systems.

(70) In one embodiment, the local server **310** may enable the gaming devices **314** within the gaming establishment **302** to participate in one or more games that share one or more progressive or pari-mutuel prizes with other gaming establishments **302** and/or gaming devices **314**. While progressive prizes are described in embodiments herein, it should be recognized that pari-mutual prizes may be substituted as desired, and vice versa. In such an embodiment, each local server **310** may be coupled to a wide area progressive (WAP) server **320** that administers the prizes. For example, the WAP server **320** receives data from each local server **310** and/or from gaming devices **314** regarding an amount wagered by each player playing the game. WAP server **320** may allocate a portion of each wager to the prizes and may communicate the current prize amounts to local servers **310** and/or to the gaming devices **314**.

(71) The gaming devices **314** may include one or more kiosks or electronic gaming machines (EGMs) (also known as “slot machines”). The gaming devices **314** may additionally or alternatively include one or more desktop computers or one or more mobile gaming devices **322**, such as, without limitation, cellular phones, tablet computing devices, and/or laptops. Mobile gaming devices **322** may connect to local server **310**, WAP server **320**, and network **308** via a wireless data network represented by cell tower **324**. For example, mobile gaming devices **322** may

connect to any suitable network **308** (and thereby to local servers **310** and/or WAP server **320**) via a "3G", "4G" or a "5G" wireless data network. It should be recognized that mobile gaming devices **322** may additionally or alternatively connect to network **308** using any suitable wireless network, such as a wireless Ethernet network. For convenience, gaming devices **314** described herein may also include mobile gaming devices **322**.

(72) One or more point-of-sale ("POS") terminals **326** or redemption kiosks may also be included within each gaming establishment **302** to enable players to "cash out" winnings from one or more gaming devices **314** and/or to perform other account management activities related to player accounts. The POS terminals **326** may be connected to the local server **310**, for example, and/or to the WAP server **320** as desired.

(73) In addition, the system **300** may include an accounting or auditing system **328** coupled to WAP server **320**, the local server **310**, and/or a gaming device **314**, for example, through network **308**. Accounting (auditing) system **328** may be used to audit and/or track components of system **300** to ensure compliance with applicable regulations.

(74) In one embodiment, a plurality of gaming devices **314** having different operating systems and/or system architectures may connect to the local server **310** or to another suitable server to play one or more games of chance. In such an embodiment, the gaming devices **314** may be used to play a session bingo game, for example, or any other game of chance.

(75) During operation, the player utilizes or selects a gaming device **314** and initiates a gaming session for playing one or more games of chance ("Games"). Optionally, the player inserts a player reward card or enters a player reward number or other identification information into gaming device **314**. If the identification information is entered, the gaming device **314** may transmit the identification information to local server **310** or accounting system **328** for authentication, or authentication may be accomplished locally within the gaming device **314**. The local server **310** communicates with player reward server **312** to establish the player's identity and to associate the gameplay with the player account. The local server **310** authenticates the player and gaming device **314** and authorizes the player to play the game or games on gaming device **314** if desired or required.

(76) When game play is initiated, during selection of the game, or during play of the game, the player may be required to purchase or generate credits. The player may purchase or generate credits by inserting cash or a ticket-in-ticket-out voucher into gaming device **314**, using cashless transfer systems, or another device. Cash, ticket-in-ticket-out vouchers, credit cards or debit cards are examples of physical items associated with the gaming device. Alternatively, or additionally, the player may transfer credits or cash to the gaming device **314** from banking accounts, credit accounts, gaming establishment accounts, and/or gaming company accounts via cashless systems. In one embodiment, computer-generated credits may be used with gaming device **314**, for example, as part of a free-to-play game.

(77) In practice, the player selects a game to play and enters a wager on the gaming device **314**. The gaming device **314** transmits data representative of the selected game and the wager to the local server **310**. If the player selects a game that is at least partially operated by the WAP server **320** or that includes one or more progressive prizes administered by WAP server **320**, local server **310** transmits the wager and game information and/or selection to WAP server **320**. The WAP server **320** may increment the progressive prizes based on the wager received from the player and may communicate the updated prize amounts via the network **308** to all other players (via associated gaming devices **314**) playing to win the progressive prizes.

(78) The player plays the game on the gaming device **314**. The following gameplay is described as being administered by the WAP server **320**. However, it should be recognized that the gameplay (i.e., the play of the game of chance) may be alternatively or additionally administered by the local server **310** and/or the gaming device **314**. For example, if the gaming device **314** is a cellular phone or a tablet computing device, the gameplay may be administered through an application installed

on the gaming device **314**.

(79) In one embodiment, the player may play a game of bingo by selecting a game or game type, one or more player cards, selecting one or more winning patterns for the player cards, and/or selecting one or more numbers or other player indicia for the player cards using the gaming device **314**. The selected player cards, winning patterns, and player indicia are transmitted to WAP server **320**. The player cards are included within one or more game tickets issued by WAP server **320**, and the game tickets are communicated to the gaming device **314** via the network **308** and the local server **310**. The WAP server **320** selects or receives randomly generated house indicia and compares the house indicia to the player indicia and the pattern or patterns selected for the player cards. Alternatively, the functions described herein (e.g., comparing the house indicia to the player indicia and the pattern or patterns selected for the player card) may be performed in the gaming device **314**. It should be recognized that the house indicia may be randomly generated using a randomization device, such as hardware, firmware, and/or software-based random number generator (RNG), a ball blower or console, a ball cage, and/or any other suitable device or machine that enables numbers or other house indicia to be randomly generated. In an alternative embodiment, the WAP server **320** (or another device) may designate a server, computer, or another device to provide randomly selected house indicia during the game and may receive the house indicia from the designated device.

(80) WAP server **320** determines whether the player wins a prize based on the comparison of the house indicia to the player indicia. For example, WAP server **320** determines whether the player indicia within the pattern or patterns selected for each card match the house indicia that were randomly determined (sometimes also referred to as the house indicia that were “called”). If the player indicia within a pattern match the called house indicia, the player may win a prize based on a pay table associated with the game. The prize may be one of the progressive prizes or the prize may be a fixed prize identified in the pay table. WAP server **320** determines the appropriate payout to be paid to the player based on the pay table and transmits data representative of the payout to local server **310**.

(81) Local server **310** receives the payout data and credits the player account accordingly. In addition, local server **310** may transmit the gameplay data and/or payout data to player reward server **312** to enable player reward server **312** to update the player history and other gameplay data for the player. When the player is done playing, the player may “cash out” some or all of the credits in the player account or may deposit the credits into the player account using POS terminal or kiosk **326**, or apply credits to a cashless system, for example. The player account may be stored on gaming device **314**, local server **310**, player reward server **312**, remote server, for example.

(82) In one embodiment, the player may enter the wager and/or may initiate play of the game on a first gaming device **314** and may complete the gameplay on a second gaming device **314**. Alternatively, the player plays the game on the first gaming device **314** and receives the results of the gameplay (e.g., whether the player won and how much the winnings are) on the second gaming device **314**. For example, the player may begin playing the game on a kiosk or electronic gaming machine and may complete the game or view the results of the game on a cell phone. In such an embodiment, the WAP server **320** and/or local server **310** may transmit the player's gameplay data from the first gaming device **314** to the second gaming device **314**.

(83) FIG. **4** is a block diagram of another system **300** that may be used to play one or more games of chance, such as a slot, bingo, keno, or any game of chance. Unless otherwise specified, the system **300** is similar to system **300** (shown in FIG. **3**) and similar components are labeled in FIG. **4** with the same reference numerals used in FIG. **3**. It should be understood that more or less components may be included within the various embodiments described herein.

(84) In the embodiment shown in FIG. **4**, the system **300** includes a plurality of gaming devices **314** that are positioned in a plurality of gaming establishments **302**. Gaming devices **314** may connect to a server **408** through a wireless access point **412**. The wireless access points **408**

includes an antenna **416** configured to wirelessly transmit to and receive signals from antennas **320** associated with the gaming devices **314**. Wireless communications systems and methods are understood by one of ordinary skill in the art and as such are not described in detail here. For example, the gaming devices **314** may be playing one or more stand alone or Internet-based games that connect to the WAP server **320** through a server **408**. In some embodiments, one or more gaming devices **314** may connect to the WAP server **320** and/or to the player reward server **312** through a wireless data network as described above. Accordingly, the gaming devices **314** interact with WAP server **320** to play the game, and WAP server **320** performs the game administration and other tasks handled by local server **310** as described above in FIG. **3**. In a similar manner, a POS terminal **326** may connect to a gaming device **314** and/or WAP server **320** via network **308**. In other respects, system **300** performs in a similar manner as described above.
(85) During operation, the player utilizes or selects a gaming device **314** and initiates a gaming session to play one or more games on the gaming device **314**. The player inserts a player reward card, enters a player reward number, provides biometric information, provides electronic identification via a cell phone or the like, or other identification information into the gaming device **314**. In some instances, additional second or third authentication means are utilized such as passwords, fingerprint scans, biometric information, or the like. The gaming device **314** transmits the identification information to player reward server **312** to establish the player's identity and to associate the gameplay with the player account. The player reward server **312** authenticates the player and the gaming device **314** and may authorize the player to play the game on the gaming device **314**. In one embodiment, the gaming device **314** also transmits the identification information to the WAP server **320** to enable the WAP server **320** to associate the player with the game to be played. As previously described, player identification or authentication may be optional.
(86) In another embodiment, the WAP server **320** authenticates the player using the player identification information in addition to, or instead of, the authentication performed by the player reward server **312**. In some embodiments, the player reward server **312** is omitted and the functions of player reward server **312** are incorporated within WAP server **320**.
(87) The player selects a game to play and enters a wager using gaming device **314**. If the player selects a game that is operated by the WAP server **320** or that includes one or more progressive prizes administered by the WAP server **320**, the gaming device **314** transmits the wager and game selection to the WAP server **320**. The WAP server **320** may increment the progressive prizes based on the wager received from the player and may communicate the updated prize amounts over the wireless channel via the server **408** to all other players (via associated gaming devices **314**) playing to win the progressive prizes.
(88) Although shown as a wireless network, it is contemplated that the same functionality may be implemented in a wired system, or any combination of both.
(89) The player plays the game on gaming device **314**. The following gameplay is described as being administered by the WAP server **320**. However, it should be recognized that the gameplay may be alternatively or additionally administered by the gaming device **314**. For example, if the gaming device **314** is a cellular phone or a tablet computing device, the gameplay may be administered through an application installed on gaming device **314**.
(90) FIG. **5** is a schematic of a computing or mobile device, or server, such as one of the devices described above, according to one exemplary embodiment. Computing device **500** is intended to represent various forms of digital computers, such as smartphones, tablets, kiosks, laptops, desktops, workstations, personal digital assistants, servers, blade servers, mainframes, and other appropriate computers. Computing device **550** is intended to represent various forms of mobile devices, such as personal digital assistants, cellular telephones, smart phones, and other similar computing devices. The components shown here, their connections and relationships, and their functions, are meant to be exemplary only, and are not meant to limit the implementations described and/or claimed in this document.

(91) Computing device **500** includes a processor **502**, memory **504**, a storage device **506**, a high-speed interface or controller **508** connecting to memory **504** and high-speed expansion ports **510**, and a low-speed interface or controller **512** connecting to low-speed bus **514** and storage device **506**. Each of the components **502**, **504**, **506**, **508**, **510**, and **512**, is interconnected using various busses, and may be mounted on a common motherboard or in other manners as appropriate. The processor **502** can process instructions for execution within the computing device **500**, including instructions stored in the memory **504** or on the storage device **506** to display graphical information for a GUI on an external input/output device, such as display **516** coupled to high-speed controller **508**. In other implementations, multiple processors and/or multiple buses may be used, as appropriate, along with multiple memories and types of memory. Also, multiple computing devices **500** may be connected, with each device providing portions of the necessary operations (e.g., as a server bank, a group of blade servers, or a multi-processor system).
(92) The memory **504** stores information within the computing device **500**. In one implementation, the memory **504** is a volatile memory unit or units. In another implementation, the memory **504** is a non-volatile memory unit or units. The memory **504** may also be another form of computer-readable medium, such as a magnetic or optical disk.
(93) The storage device **506** is capable of providing mass storage for the computing device **500**. In one implementation, the storage device **506** may be or contain a computer-readable medium, such as a hard disk device, an optical disk device, or a tape device, a flash memory or other similar solid-state memory device, or an array of devices, including devices in a storage area network or other configurations. A computer program product can be tangibly embodied in an information carrier. The computer program product may also contain instructions that, when executed, perform one or more methods, such as those described above. The information carrier is a computer- or machine-readable medium, such as the memory **504**, the storage device **506**, or memory on processor **502**.
(94) The high-speed controller **508** manages bandwidth-intensive operations for the computing device **500**, while the low-speed controller **512** manages lower bandwidth-intensive operations. Such allocation of functions is exemplary only. In one implementation, the high-speed controller **508** is coupled to memory **504**, display **516** (e.g., through a graphics processor or accelerator), and to high-speed expansion ports **510**, which may accept various expansion cards (not shown). In the implementation, low-speed controller **512** is coupled to storage device **506** and low-speed bus **514**. The low-speed bus **514**, which may include various communication ports (e.g., USB, Bluetooth, Ethernet, wireless Ethernet or the like) may be coupled to one or more input/output devices, such as a keyboard, a pointing device, a scanner, or a networking device such as a switch or router, e.g., through a network adapter.
(95) The computing device **500** may be implemented in a number of different forms, as shown in the figure. For example, it may be implemented as a standard server **520**, or multiple times in a group of such servers. It may also be implemented as part of a rack server system **524**. In addition, it may be implemented in a personal computer such as a laptop computer **522**. Alternatively, components from computing device **500** may be combined with other components in a mobile device (not shown), such as device **550**. Each of such devices may contain one or more of computing device **500**, **550**, and an entire system may be made up of multiple computing devices **500**, **550** communicating with each other.
(96) Computing device **550** includes a processor **552**, memory **564**, an input/output device such as a display **554**, a communication interface **566**, and a transceiver **568**, among other components. The device **550** may also be provided with a storage device, such as a micro-drive or other device, to provide additional storage. Each of the components **550**, **552**, **564**, **554**, **566**, and **568**, are interconnected using various buses or similar, and several of the components may be mounted on a common motherboard or in other manners as appropriate.
(97) The processor **552** can execute instructions within the computing device **550**, including

instructions stored in the memory **564**. The processor may be implemented as a chipset of chips that include separate and multiple analog and digital processors. The processor may provide, for example, for coordination of the other components of the device **550**, such as control of user interfaces, applications run by device **550**, and wireless communication by device **550**.

(98) Processor **552** may communicate with a user through control interface **558** and display interface **556** coupled to a display **554**. The display **554** may be, for example, a TFT LCD (Thin-Film-Transistor Liquid Crystal Display) or an OLED (Organic Light Emitting Diode) display, or a QLED (Quantum Light-Emitting Diode), or other appropriate display technology. The display interface **556** may comprise appropriate circuitry for driving the display **554** to present graphical and other information to a user. The control interface **558** may receive commands from a user and convert them for submission to the processor **552**. In addition, an external interface **562** may be provide in communication with processor **552**, to enable near area communication of device **550** with other devices. External interface **562** may provide, for example, for wired communication in some implementations, or for wireless communication in other implementations, and multiple interfaces may also be used.

(99) The memory **564** stores information within the computing device **550**. The memory **564** can be implemented as one or more of a computer-readable medium or media, a volatile memory unit or units, or a non-volatile memory unit or units. Expansion memory **574** may also be provided and connected to device **550** through expansion interface **572**, which may include, for example, a SIMM (Single In Line Memory Module) card interface. Such expansion memory **574** may provide extra storage space for device **550** or may also store applications or other information for device **550**. Specifically, expansion memory **574** may include instructions to carry out or supplement the processes described above and may include secure information also. Thus, for example, expansion memory **574** may be provide as a security module for device **550** and may be programmed with instructions that permit secure use of device **550**. In addition, secure applications may be provided via the SIMM cards, along with additional information, such as placing identifying information on the SIMM card in a non-hackable manner.

(100) The memory may include, for example, flash memory and/or NVRAM memory, as discussed below. In one implementation, a computer program product is tangibly embodied in an information carrier. The computer program product contains instructions that, when executed, perform one or more methods, such as those described above. The information carrier is a computer- or machine-readable medium, such as the memory **564**, expansion memory **574**, or memory on processor **552**, that may be received, for example, over transceiver **568** or external interface **562**.

(101) Device **550** may communicate wirelessly through communication interface **566**, which may include digital signal processing circuitry where necessary. Communication interface **566** may provide for communications under various modes or protocols, such as GSM voice calls, SMS, EMS, or MMS messaging, CDMA, TDMA, PDC, WCDMA, CDMA2000, or GPRS, among others. Such communication may occur, for example, through radio-frequency transceiver **568**. In addition, short-range communication may occur, such as using a Bluetooth, Wi-Fi, or other such transceiver (not shown). In addition, GPS (Global Positioning system) receiver module **570** may provide additional navigation- and location-related wireless data to device **550**, which may be used as appropriate by applications running on device **550**.

(102) Device **550** may also communicate audibly using audio codec **560**, which may receive spoken information from a user and convert it to usable digital information. Audio codec **560** may likewise generate audible sound for a user, such as through a speaker, e.g., in a handset of device **550**. Such sound may include sound from voice telephone calls, may include recorded sound (e.g., voice messages, music files, etc.) and may also include sound generated by applications operating on device **550**.

(103) The computing device **550** may be implemented in a number of different forms, as shown in the figure. For example, it may be implemented as a cellular telephone **560**. It may also be

implemented as part of a smart phone **582**, personal digital assistant, a computer tablet, or other similar mobile device.

(104) Thus, various implementations of the systems and techniques described here can be realized in digital electronic circuitry, integrated circuitry, specially designed ASICs (application specific integrated circuits), computer hardware, firmware, software, and/or combinations thereof. These various implementations can include implementation in one or more computer programs that are executable and/or interpretable on a programmable system including at least one programmable processor, which may be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, at least one input device, and at least one output device.

(105) These computer programs (also known as programs, software, software applications or code) include machine instructions for a programmable processor, and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the terms "machine-readable medium" "computer-readable medium" refers to any computer program product, apparatus and/or device (e.g., magnetic discs, optical disks, memory, Programmable Logic Devices ("PLDs") used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The term "machine-readable signal" refers to any signal used to provide machine instructions and/or data to a programmable processor.

(106) To provide for interaction with a user, the systems and techniques described here can be implemented on a computer having a display device (e.g., a CRT (cathode ray tube), LCD (liquid crystal display), or other monitor types) for displaying information to the user and a keyboard and a pointing device (e.g., a mouse, joy stick, trackball, or similar device) by which the user can provide input to the computer. Other kinds of devices can be used to provide for interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback (e.g., visual feedback, auditory feedback, or tactile feedback); and input from the user can be received in any form, including acoustic, speech, or tactile input.

(107) The systems and techniques described here can be implemented in a computing system (e.g., computing device **500** and/or **550**) that includes a back end component (e.g., as a data server, slot accounting system, player tracking system, or similar), or that includes a middleware component (e.g., an application server), or that includes a front end component (e.g., a client computer having a graphical user interface or a Web browser through which a user can interact with an implementation of the systems and techniques described here), or any combination of such back end, middleware, or front end components. The components of the system can be interconnected by any form or medium of digital data communication (e.g., a communication network). Examples of communication networks include a local area network ("LAN"), a wide area network ("WAN"), and the Internet.

(108) The computing system can include clients and servers. A client and server are generally remote from each other and typically interact through a communication network. The relationship of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship to each other.

(109) FIG. **6** is a block diagram of a gaming device **114** that may be used with system **300** (shown in FIG. **3**) or system **400** (shown in FIG. **4**). As described above, the gaming device **114** is a computing device **200** (such as an EGM) that includes a plurality of computing device components **602** positioned within or on a cabinet or other housing. In one embodiment, computing device component manager or processor **640** include first display **116** and second display **118**. In addition, gaming device **114** may include a plurality of gaming device components **602** including a bill acceptor or bill validator **604**, a card reader **606**, a barcode scanner **608**, a printer **610**, an intrusion detection system **612**, a randomization device **614** (such as an RNG), and an accounting interface **616** that are positioned within, or coupled to, the cabinet or housing of the gaming device. In one

embodiment, gaming device **114** may also include at least one lighting element **618** coupled to the cabinet or housing.
(110) It should be recognized that in some embodiments, a gaming device **114** may not include each gaming device component **602** illustrated in FIG. **6**. For example, if the gaming device **114** is a cellular phone or a tablet, the gaming device may not include bill acceptor **604**, card reader **606**, barcode scanner **608**, and/or printer **610**. Rather, in some embodiments, the functions of each omitted gaming device component may be replaced by equivalent software, hardware, and/or firmware if desired. Optional components may be designated using dashed lines in the figures.
(111) The bill acceptor **604** is a payment input device that enables gaming device **114** to receive and identify paper currency, ticket-in-ticket-out vouchers, or other physical items representing a monetary value. For example, bill acceptor **604** may receive and identify dollar bills or other currency that are inserted into bill acceptor **604**. In one embodiment, bill acceptor **604** includes a scanner that scans paper currency inserted therein. The bill acceptor **604** may also include optical character recognition (OCR) capabilities that enable bill acceptor **604** to identify the amount of currency inserted into bill acceptor **604** from a scanned image of the currency. The bill acceptor **604** may transmit data representative of the amount of currency inserted into gaming device **114** to controller or processor **640**, for example. The controller or processor **640** may cause the amount of currency to be converted into credits usable with the game and may add the credits to the player's account. Those skilled in the art will recognize that similar principals apply to cashless wagering systems.
(112) The card reader **606** is a device that “reads,” or obtains data encoded in, player reward cards or other cards or media that are inserted into reader. In one embodiment, the card reader **606** is a magnetic or optical card reader that reads barcodes or magnetic strips included within a player reward card. In another embodiment, the card reader **606** wirelessly reads data encoded within the player reward card by accessing a chip, such as a radio frequency identification (“RFID”) chip, embedded within the card or other similar authentication means. The card reader **606** reads the data obtained from the cards and transmits the data to the computing device components/processor **640**. In one embodiment, the card reader **606** is used to read player identification information encoded within player reward cards. Those skilled in the art will recognize various means of player identification as herein described. The controller or processor **640** may transmit the player identification information to player reward server or other external component to identify the player, track past or present player activity, to allow for the transfer of funds or credits, to facilitate authenticating the player, and/or to authorize the player to play a game on gaming device **114**. Those skilled in the art will recognize that many alternate means of electronic funds transfer may be employed in a cashless system and may include near field communication, Bluetooth communication, wireless communication, biometric identifications, etc., which may be associated with bank accounts, credit card accounts, gaming related accounts, player accounts, debit accounts, etc. Even in a cashless system, a bill acceptor or similar, accepting a physical item such as currency or ticket-in-ticket-out vouches or similar may also be employed for those not utilizing the cashless system. In one embodiment, the player may “log in” to the gaming device **114** by swiping the player reward card or otherwise passing the player reward card through or inserting the player reward card into the card reader **606**. In another embodiment, the player may enter a number or other identifier associated with the player reward card into the gaming device **114**, through the user interface devices for example, instead of using the card reader **606**. In another embodiment, the insertion of the player reward card and player entering the identifier into user interface device may be combined. In yet another embodiment, the player may use a near field communication (“NFC”) device to read the player reward card or data representative of the player card. Alternatively, the player reward card may be associated with an application on a cell phone or tablet which wirelessly communicates with the card reader or similar system.
(113) In one embodiment, the barcode scanner **608** is an optical or a magnetic scanner that is

optimized to read barcodes on media positioned proximate to the scanner and may also include RFID sensors, blue tooth connectivity, near field communications devices, etc. For example, the barcode scanner **608** may be optimized to read barcodes printed on paper receipts (sometimes referred to as “tickets” or vouchers, not to be confused with game or player tickets that may include player selected patterns, player indicia, and the like) and/or barcodes displayed electronically on a cell phone or tablet computing device. It should be recognized that the barcodes read by the barcode scanner **608** may be linear or one-dimensional barcodes, two-dimensional barcodes, or may even include data represented in a form other than a barcode. For example, the barcode scanner **608** may read images and/or text indicative of data, such as currency or credits, usable with gaming device **114**. The barcode scanner **608** extracts the data from the barcode and transmits the data to controller/processor **640**. For example, the barcode scanner **608** may scan a paper receipt or voucher that includes an amount of currency or credits usable by the player with a gaming device **114** and may transmit the amount of credits to the controller/processor **640**. In such an example, the barcode scanner **608** may act as a payment input device. The controller/processor **640** may cause the amount of currency or credits to be displayed to the player on first display **616** (or on any display) to inform the player how many credits or currency is available to be used in playing a game.

(114) The printer **610** may be used to print paper receipts (also known as tickets as described above), ticket-in-ticket-out vouchers, or other physical items representing a monetary value that indicate an amount of currency or credits available to the player, player comps, and the like. In many locations, the tickets or receipts may alternatively be referred to as vouchers. The printer **610** may act as a payment output device that enables a player to cash out or withdraw money or credits from the gaming device **114** by printing a voucher representative of the money or credits. In one embodiment, the printer **610** is a thermal printer that is fed by a roll of paper or any suitable paper stock. In a further embodiment, the roll of paper includes one or more watermarks that are visible when the printer **610** has printed the receipt on the paper. Alternatively, the printer **610** may print the watermark on the receipt, or may include another security mechanism to facilitate preventing counterfeit receipts from being made. For example, the printer **610** may include an image or a code on the receipt that identifies the gaming device **114**, the printer **610**, or another component of the gaming device along with a time that the receipt was printed, serial number, date, location, or other desired information. Other suitable security mechanisms may be used as well. It should be recognized that the barcode scanner **608** and the printer **610** may cooperate such that a security mechanism printed on the receipt may be received and validated by the barcode scanner, in conjunction with controller/processor **640**, for example. The barcode scanner **608** may be located remotely from the gaming device **114**, such as within a redemption kiosk, a casino cage, or the like.

(115) The intrusion detection system **612** notifies the controller/processor **640** if a case, cabinet, or other housing enclosing components of the gaming device **114** is opened or modified without authorization. In one embodiment, the intrusion detection system **612** includes a pair of contacts that may be physical, magnetic, optical, or similar that transmit an electronic signal to the controller/processor **640** if the housing of the gaming device **114** is opened (e.g., if the opening of the housing separates the contacts). In another embodiment, the intrusion detection system **612** may include a light sensor that detects a change in the light within the housing of the gaming device **114**. The intrusion detection system **612** may also include a key or another mechanism for disabling the operation of the game or transmission of the signal to the controller/processor **640** in the event that maintenance or other authorized or unauthorized access to the gaming device **114** components is desired or occurs.

(116) In one embodiment, the intrusion detection system **612** includes a software program (a “monitoring program”) that monitors one or more applications installed on the gaming device **114**. For example, if the gaming device **114** is a cell phone that includes an application for playing the game thereon, the monitoring program may monitor the application to determine whether the

application is modified without authorization. In one embodiment, the monitoring program stores a hash value or a digital fingerprint of the application when the application is installed and/or when the application undergoes authorized modification (e.g., if the application is updated or patched). However, if the monitoring program determines that the application has been modified without authorization, the monitoring program may cause a signal or another notification to be transmitted to the controller/processor **640**. For example, the monitoring program may periodically calculate a new hash value of the application and/or create a new digital fingerprint of the application. The monitoring program then compares the new hash value and/or digital fingerprint to the stored hash value and/or digital fingerprint. If the hash values or fingerprints are different, the monitoring program may determine that the application has been modified without authorization. It should be understood that the hash value, the monitoring program, and/or the digital fingerprint may be generated by any suitable means and may be encrypted for additional security.

(117) In response to the signal or notification from the intrusion detection system **612** and/or the modification program, the controller/processor **640** may perform one or more actions. For example, the controller/processor **640** may alert an administrator within gaming establishment by transmitting a message via communication device, may cause audio output device to emit an alarm or another audible alert, may cause a display **116**, **118** to display an error or a warning, message, and/or may disable the application and/or the gaming device **114** such that the game is unable to be played on the gaming device.

(118) In one embodiment, the randomization device is an electronic random number generator (“RNG”) or pseudo random number generator (“PRNG”) **614** or a permutation generator that may be implemented by a dedicated hardware device with associated embedded software. Electronic random number generators or pseudo random number generators are used interchangeably herein. Alternatively, the RNG **614** or the permutation generator may be implemented entirely in software executing on gaming device **114**. The RNG **614** may be used to randomly determine a game outcome for the game of chance. In one embodiment, the RNG **614** or the permutation generator provides house or game draws of between 1 and n numbers, where n may be a suitable number based on the game type selected to be played by the player. The RNG **614** or the permutation generator may be programmed via hardware, software, or firmware to provide a particular range of numbers (or other indicia) and numbers of draws for a particular application. For example, in one embodiment of bingo according to the present disclosure, the RNG **614** or the permutation generator initially provides 24 randomly generated numbers having values between 1 and 75 for each game. In other embodiment other methods or numeric values may be used. Additional draws or numbers may be provided to play the game to conclusion depending on the particular implementation as described in greater detail herein. In addition, the RNG **614** or the permutation generator may be used to randomly select a plurality of player indicia to be used with one or more player cards. In embodiments in which a processor, such as controller/processor **640**, is described as randomly selecting indicia, it should be recognized that controller/processor may interface with randomization device **614** or the permutation generator to select the indicia. In other embodiments, controller/processor **640** may include randomization device **614** or the permutation generator, or may execute instructions to perform the functions of randomization device **614** or the permutation generator.

(119) The accounting interface **616** is used to interface with an accounting system, such as a slot accounting system, at or operated by a gaming establishment. Accounting interface **616** may include or be connected to a network interface, such as the communication device **308** for use in communicating gameplay data, player identification information, and/or other data to the accounting system for accounting and/or auditing purposes.

(120) The lighting element **618** may include, for example, one or more LEDs, slot machine candles, fluorescent tubes, and/or any other element that emits light as controlled or directed by the controller/processor **640**. In one embodiment, the lighting element **618** is activated to display light,

or one or more lighting patterns, when the controller/processor **640** determines that a winning ticket was scanned via the card reader **606** or when the controller/processor otherwise determines that a ticket is a winning ticket. The lighting elements **618** may also be activated upon receipt of a signal from the intrusion detection system **612** (e.g., upon the determination that the gaming device **114** has been opened and/or modified without authorization) and/or upon any other suitable determination.

(121) In one embodiment in which the gaming device **114** or kiosk may interface with another gaming device operated by or otherwise associated with the player, such as a cell phone, tablet, or another mobile device. For example, the gaming machine or kiosk may be configured to transmit a result of one or more games of chance to the player's mobile device to notify the player whether one or more player cards or game tickets are winning cards or tickets.

(122) FIG. **7** is a block diagram of a plurality of program modules **700** that may be used with the systems shown and described herein to administer one or more games of chance. In one embodiment, one or more program modules **700** are installed and/or stored within local server, WAP server, and/or gaming devices. For example, program modules **700** may be stored in memory device of local server, WAP server, and/or gaming devices.

(123) The program modules **700** are hardware, firmware, or software programs or applications that, when executed by a processor, cause the processor to perform the functions described herein. In one embodiment, the program modules **700** include a wrapper program module **702**, a plurality of game modules **704**, a pay table module **706**, a progressive prize module **708**, a local prize module **710**, a flashboard module **712**, and/or an accounting module **713**. A first plurality **714** of the program modules **700** may be installed within each local server and/or WAP server and a second plurality **716** of the program modules **800** may be installed within each gaming device. It should be recognized that in embodiments in which the game of chance is administered by gaming device (e.g., when a cell phone or a tablet computing device is used as gaming device), some or all of the first plurality **714** of program modules **700** may be incorporated within gaming device and executed by a processor of a gaming device. Alternatively, some or all of the second plurality **716** of the program modules **700** may be incorporated within a local server and/or WAP server. Together, the wrapper program module **702**, the game modules **704**, and the other program modules **700** that present and/or administer one or more games may be referred to herein as a game application, or an application.

(124) In one embodiment, the wrapper program module **702** is used at least in part to provide a graphical user interface ("GUI") on a first display of the gaming device. The wrapper program module **702** operates to provide an entry point or a game entry interface for a player to access the gaming device, and to enable the player to select a game of chance to be played on the gaming device. For example, the games of chance may be categorized into a plurality of game sizes and a plurality of game variations. The wrapper program module **702** may present the game sizes and the game variations to the player, using a display, and may enable the player to select a game to play by selecting a game size and game variation through user interface device.

(125) In one embodiment, the wrapper program module **702** may present a list of games or game variations to the player for selection on a display. If the player selects a size and variation, wrapper program module **702** calls or branches to a game module **704** that provides the selected game and variation.

(126) In one embodiment, the game modules **704** each provide a game associated with the selected game size and/or game variation to the player using gaming device, local server, and/or WAP server. Accordingly, in one embodiment, each game is provided by a separate game module **704**. Alternatively, each game module **704** may provide more than one game to the player.

(127) The pay table module **706** provides a pay table associated with each game such that one or more pay tables may be associated with each game module **704**. In one embodiment, the pay table module **706** provides a pay table associated with a game when the game module **704** requests the

pay table and/or when a predetermined event occurs during the game. The pay tables associated with a game may be changed as desired by a game operator by any suitable means. The predetermined event may include, for example, the player selecting a “See Pays” or another icon displayed on the display that represents a request to view the pay table for the game. The predetermined event may also include reaching a point in the game in which the house indicia are matched to the player indicia within a selected pattern to determine whether the player wins a prize.
(128) The progressive prize module **708** may be used to administer aspects of one or more progressive prizes, such as one or more progressive prizes offered to players playing across network. For example, the progressive prize module **708** may receive information regarding an amount wagered by each player playing a game that has a chance to win the progressive prize. The progressive prize module **708** may allocate a first portion of each wager to a first progressive prize to increase the size of the progressive prize. The progressive prize module **708** may allocate a second portion of each wager to a second progressive prize, and may continue in a similar manner for any additional progressive prizes, if desired or applicable. Accordingly, a plurality of progressive prizes may be provided for each game and may be at least partially funded by each or selected wagers.
(129) The local prize module **710** may be used to administer aspects of one or more local prizes, such as one or more prizes that may be won by players playing against each other within a gaming establishment. In addition, the local prize module **710** may administer aspects of one or more fixed prizes, such as prizes that may be won only by individual players playing on respective gaming device. Accordingly, fixed or individual prizes may be awarded to a player based on the gameplay of the player relative to a randomization device of gaming device, rather than based on winning against other players.
(130) In one embodiment, the slot module **712** may be used to control and conduct slot games in the manner and for the purposes detailed below.
(131) The accounting module **713** may be used to interface with an accounting system, such as a slot accounting system or auditing system, at or operated by a gaming establishment. In one embodiment, the accounting module **713** is incorporated within, or executed by, accounting interface. Any suitable data, such as gameplay data, player identification information, prizes won by a player, and/or any other suitable data may be collected and transmitted by the accounting module **713**.
(132) It should be recognized that two or more program modules **700** may be combined together such that the functionality of each program module **700** is incorporated into the combined module. Likewise, each program module **700** may be split into two or more sub-modules that each perform a portion of the functionality of the program module **700** being split. Accordingly, while the above-described program modules **700** are described individually, each may be combined or split into other sub-modules as desired.
(133) FIG. **8** is an illustration of one embodiment of a select screen **800** according to the embodiments of the present invention. In one embodiment, select screen **800** is comprised of a touch screen LCD display **802**. Those skilled in the art will recognize that much of the functionality described in FIG. **8** through FIG. **12** may also exist on other displays, be implemented with physical buttons or the like, or any combination thereof. Moreover, while preferable, common elements may exist from table game to table game, other elements and interfaces may change, table game to table game.
(134) Game select screen **800** is captioned “Select Game” **804** which allows a player to select one of a plurality of table games. As illustrated, a player may select from the table games of baccarat **824**, blackjack **830**, roulette **826** or craps **828**. While these particular table games are illustrated, any number and type of table games may be offered. As discussed, certain common interface elements are included in all primary displays for convenience for the player but are not mandatory. Such elements may include a credit display **806**, a bet or wager display **808**, a menu button **810** and

a win display **812**. Additionally, wager amounts are made by utilizing the virtual chips illustrated. The casino operator may determine and set bet or wager amounts as they desire. As an example, the interface illustrates a 250 chip **814**, a $1 chip **816**, a $5 chip **818** and a $25 chip **820**. The display **800** also indicates the minimum bet or wager **822** a player may make. As betting or wagering does not occur during the select game **804** process, this functionality is not necessary for this particular display but is necessary for actual operable game screens. Once a player selects the table game they wish to play, this display is replaced by the primary game screen relative to the selected game of choice. In an alternative embodiment, the game select screen **800** includes a shuffle input (not shown) which causes the processor of the EGM to randomly cause any of the stored virtual games to be presented to the player. Such a random selection may be made after each hand or round is played or may be made after a pre-established number of hands or rounds (e.g., five) of a selected game. In one embodiment, the player may select how many hands or rounds are played before the processor randomly selects a new game. Such an embodiment adds excitement and anticipation to the player experience. In one embodiment, the player may place a bet or wager on which game the processor will next select.

(135) FIG. **9**A illustrates one embodiment of a user interface **900** of a touch screen LCD button deck **902** of the electronic table game of craps of the present invention. Although the user interface **900** is utilized for the table game of craps, it may be utilized for other table games where appropriate. A number of common elements may exist between various table game user interfaces, some elements may be common between such user interfaces. Such common elements may include a service button **904** used to alert the casino staff there may be a problem with the EGM, a collect button **906** which allows a player to collect or cash out their current credits generally by way of a printed credit voucher, a change game button **908** which allows a player to change games by toggling the display to the select screen shown in FIG. **8** or similar, and a help/pays button **910** which provides the player with game rules, game information, paytable and the like. For the particular game of craps or similar games, a plurality of buttons are displayed which allow the player to clear a bet **912**, undo a previous operation **914**, or initiate the craps game by touching the roll button **916**.

(136) FIG. **9**B illustrates one embodiment of the top display **920** of an EGM relating to the electronic table game of craps according to the embodiments of the present invention. Those skilled in the art will recognize that any table game may be offered on a variety of display configurations including a single landscape mode display or a single portrait mode display with a separate button deck as illustrated in FIG. **9**A, a single landscape mode display or a single portrait mode display with button deck functionality included in the display, dual displays as illustrated herein, dual displays with the button deck functionality included in one of the two displays, or three displays, generally with landscape orientation, with either a separate button deck or button deck functionality included within one of the three displays.

(137) In the case of FIG. **9**B, one embodiment of the top display **920** is comprised of either a touch screen display or non-touch screen display **922**. Generally, the top display **922** either represents an image of the table game being played, a logo of the game being played, a virtual dealer, or other images. As illustrated, the image displayed on top display **922** is a virtual craps table layout **924** which includes a number of betting positions **926**. Once the player places one or more bets or wagers, the player may initiate the games by touching the roll button **916** as shown in FIG. **9**A, wherein the virtual dice **928** are thrown onto the craps table layout **924**. In one embodiment, this is an animated sequence to emulate real dice where the dice travel and rotate until they eventually come to a resting position. The dice indicia, as represented by dots or pips, represent numbers which are added together to determine the total number of dots or pips on the top surface of the dice **928**. As shown, the dice **928** together show a total value of 9.

(138) FIG. **9**C illustrates one embodiment of the bottom display **940** of an EGM relating to the electronic table game of craps according to the embodiments of the present invention. Bottom

display **940** is comprised, in one embodiment, of a touch screen display **942** displaying a craps game layout **941**. Bottom display **940** also includes a number of generally common elements which may include a credit display **946**, a bet or wager display **948**, a menu button **950** and a win display **952**. In any table game where no decisions are required of the player, such as craps, roulette, baccarat, or similar, a repeat button may be included to allow for repeating the previous bet(s) or wager(s). The repeat button may also serve to allow for automated continual play where once a player holds the repeat bet button down for a predetermined time or similar, the electronic table game will automatically play additional games with the same betting or wagering scheme until the player again presses the play or repeat button or similar prescribed player input to end the automated session. Additionally, wager amounts are made by utilizing the virtual chips illustrated. The casino operator may determine and set bet or wager amounts as they desire. As an example, the interface illustrates a 250 chip **954**, a $1 chip **956**, a $5 chip **958** and a $25 chip **960**. However, a particular casino operator may configure the betting range 50, 250, 500 and $1 while another operator may configure the betting range as $5, $25, $100, and $500. There are few limitations to how an operator may configure a betting range. Those skilled in the art will recognize that although four virtual chips of various denominations are illustrated, any number of virtual chips with varying denominations may be utilized. Bottom display **940** also indicates the minimum bet or wager **962** a player may make to initiate a game.

(139) The craps layout **941** preferably includes a number of betting or wagering positions **964**. Generally, the game of craps includes a great many betting or wagering positions, each offering a different type bet or wager with some offering the same payouts with some with differing payouts. The payouts generally escalate as the probability of winning decreases. In addition, after an initial roll when a player does not roll a 7 or 11 to win or 2, 3, or 12 to lose, the player will establish their point which is the total shown on the dice on the initial roll. Pass and come line bets may allow for odds bets to increase the player bet or wager and reduce the house advantage once the point is established. This betting or wagering scheme may be configurable by the operator to allow for no odds bets, 1× odds bet, 2× odds bet, 3× odds bets, etc. Generally, the maximum odds bets will be under 50× odds bets.

(140) Initially, when a new game begins, a player places a bet or wager by using hand gestures between the virtual chips and the desired betting position **964** on the craps layout **941**. For instance, a player may touch a desired chip **954**, **956**, **958**, or **960** and slide it to the desired bet or wager position **964** or touch a desired chip **954**, **956**, **958**, or **960** and then touch the desired betting or wagering position **964**. Those skilled in the art will recognize that a variety of hand gestures may be utilized in this regard. As illustrated, a particular player has made a number of different bets or wagers as shown by virtual chips located on the craps layout **941**. The bets or wagers shown include a $25 bet or wager **966** on the pass line, a $1 bet or wager **966** on the 4 and 5, a $5 bet or wager **966** on hard way 10, and a $5 bet or wager **966** on 12. An operator may configure the game to allow for either a limited number of bets or wagers or an unlimited number of bets or wagers. This may be configured as the number of bets or wagers or by setting monetary minimum and maximum bets or wagers. Any number of bets or wagers may be placed on a betting or wagering position **964**. As an example, a player may place a $1 virtual chip **954** on the come line bet and then add a 250 virtual chip **954** to the come line bet, followed by a $25 virtual chip **960** and a $5 virtual chip **958** to the come line bet for a total bet or wager of $31.25. As a player places their bets or wagers, the bet or wager amount is deducted from their credit balance as shown in credit display **946**. The type and number of such bet or wager placements are generally only limited by the operator's desired configuration. In the event a player places the virtual chip **954**, **956**, **958**, or **960** on a betting or wagering position which is partially on one bet or wagering position and partially on another bet or wagering position, which may occur in a physical game, the game state engine of the particular game, as shown in FIG. **13**, may automatically shift the virtual chip **954**, **956**, **958**, or **960** to the position mostly covered by the virtual chip **954**, **956**, **958**, or **960** to avoid any doubt of

where the actual bet or wager resides.

(141) Once a player has concluded placing their bets or wagers, the player initiates the game by touching the roll button **916** on display **900** whereas two virtual dice **928** are rolled. In one embodiment, virtual dice **928** are animated to emulate real dice and when they come to rest, the final value of the dice spots or pips shown on the top surface of the dice **928** are added together to determine the final value. Following the dice coming to rest, the processor **1302**, shown in FIG. **13**, determines the final results and settles all bets or wagers. The final win settlement is shown in the win display **952** and then added to the current credits as shown in display **946**. In the game of craps, many bets may carry over from roll to roll if a player has not made their point for a win or rolled a 7 to end their roll session. However, when a 7 is finally rolled, generally all bets or wagers are concluded and settled with some such as come line bets or wagers winning and other bets or wagers losing.

(142) An important feature of EGMs is the number of plays possible within a given time frame, e.g., plays per hour. For some games such as blackjack or roulette, there is slight lag between the time a player initiates the game and the conclusion of the game. However, for games which include virtual physical items to determine game outcomes such as the dice in the game of craps or the roulette wheel in the game of roulette, significant time is taken to allow for the animation of the dice coming to rest in the game of craps or the roulette wheel coming to rest and the roulette ball falling into a final position. Embodiments of the present invention allow for a player to touch either the roll button for craps or the spin button in roulette to immediately advance to the final outcome by truncating much or all of the game animation thereby speeding up the game or plays per hour by a considerable amount. Any embodiments of the present invention may similarly include such speed play functionality. In one embodiment, animation and/or speed play software modules facilitate the animation and speed play functionality.

(143) FIG. **10**A illustrates one embodiment of a user interface **1000** of a touch screen LCD button deck **1002** of the electronic table game of baccarat according to the embodiments of the present invention. Although the user interface **1000** is utilized for the table game of baccarat, it may be utilized for other table games where appropriate. A number of common elements may exist between various table game user interfaces, some elements may be common between such user interfaces. Such common elements may include a service button **1004** used to alert the casino staff there may be a problem with the EGM, a collect button **1006** which allows a player to collect or cash out their current credits generally by way of a printed credit voucher, a change game button **1008** which allows a player to change games by toggling the display to the select screen shown in FIG. **8** or similar, and a help/pays button **1010** which provides the player with game rules, game information, paytable and the like. For the particular game of baccarat or similar games, a plurality of buttons are displayed allowing the player to clear a bet **1012** or initiate the baccarat game by touching the deal button **1014**.

(144) FIG. **10**B illustrates one embodiment of the top display **1020** of an EGM relating to the electronic table game of baccarat according to the embodiments of the present invention. Those skilled in the art will recognize that any table game may be offered on a variety of display configurations including a single landscape mode display or a single portrait mode display with a separate button deck as illustrated in FIG. **10**A, a single landscape mode display or a single portrait mode display with button deck functionality included in the display, dual displays as illustrated herein, dual displays with the button deck functionality included in one of the two displays, or three displays, generally with landscape orientation, with either a separate button deck or button deck functionality included within one of the three display.

(145) In the case of FIG. **10**B, one embodiment of the top display **1020** is comprised of either a touch screen display or non-touch screen display **1022**. Generally, the top display **1020** either represents an image of the table game being played, a logo of the game being played, a virtual dealer, or other images.

(146) FIG. **10**C illustrates one embodiment of the bottom display **1040** of an EGM relating to the electronic table game of baccarat according to the embodiments of the present invention. In one embodiment, bottom display **1040** is comprised of a touch screen display **1042** displaying a baccarat game layout **1044**. Bottom display **1040** also includes a number of generally common elements which may include a credit display **1046**, a bet or wager display **1048**, a menu button **1050** and a win display **1052**. Additionally, wager amounts are made by utilizing the virtual chips illustrated. The casino operator may determine and set bet or wager amounts as they desire. As an example, the interface illustrates a 250 chip **1054**, a $1 chip **1056**, a $5 chip **1058** and a $25 chip **1060**. However, a first casino operator may configure the betting range 50, 250, 500 and $1 while another operator may configure the betting range as $5, $25, $100, and $500. There are few limitations to how an operator may configure a betting range. Those skilled in the art will recognize that although four virtual chips of various denominations are illustrated, any number of virtual chips with varying denominations may be utilized. Bottom display **1040** also indicates the minimum bet or wager **1062** a player may make to initiate a game.

(147) The baccarat game layout **1044** includes a number of betting or wagering positions **1072**, **1072** and **1076**. Generally, the game of baccarat includes several betting or wagering positions, each offering a different type bet or wager. In one embodiment, the baccarat game includes a side bet known as a “Lucky 99” side bet **1078**. One skilled in the art will recognize many different type side bets may be added to the base baccarat game. In certain games, such as baccarat, side bets may be very important as they generally have a much higher hold percentage than the base game and generate a much higher win for the casino operator. For instance, the hold percentage of a player placing a wager on just the banker or player is slightly above 1%. However, if the player also places a bet on the “Lucky 99” side bet **1078**, the hold percentage may increase to over 6%, depending on the amount of the “Lucky 99” side bet **1078**. Side bets often offer higher payout multiples to attract players as shown in paytable **1082**. Side bets may become of lesser importance on some table games as the games either have higher base hold percentages or offer various traditional bets with higher hold percentages, such as the game of craps.

(148) Initially, when a new baccarat game begins, a player places a bet or wager by using hand gestures between the virtual chips and the desired betting position **1072**, **1074** and **1076** on the baccarat game layout **1044**. For instance, a player may touch a desired chip **1054**, **1056**, **1058**, or **1060** and slide it to the desired bet or wager position **1072**, **1072** and **1076** or touch a desired chip **1054**, **1056**, **1058**, or **1060**, lift the finger, and then touch the desired betting or wagering position **1072**, **1074** and **1076**. Those skilled in the art will recognize that a variety of hand gestures may be utilized in this regard. As illustrated, a player has made a number of different bets or wagers as shown by virtual chips located on the baccarat game layout **1044**. The bets or wagers shown include a $25 bet or wager **1084** on the banker bet and $5 bet or wager **1080** on the “Lucky 99” side bet **1078**. Generally, the game of baccarat only allows for a limited number of bets or wagers. Betting or wagering limitations may be configured as the number of bets or wagers or by setting monetary minimum and maximum bets or wagers. Any number of bets or wagers may be placed on a betting or wagering position. As an example, a player may place a $1 virtual chip **1056** on the banker bet and then add three 250 virtual chips **1054** to the banker bet, followed by two $25 virtual chips **1060** and a $5 virtual chip **1058** to the banker bet for a total bet or wager of $**56.75**. As a player places their bets or wagers, the bet or wager amount is deducted from their credit balance as shown in credit display **1046**. The type and number of such bet or wager placements are generally only limited by the operator's desired configuration. In the event a player places the virtual chip **1054**, **1056**, **1058**, or **1060** on a betting or wagering position which is partially on one bet or wagering position and partially on another bet or wagering position, which also occurs in a physical game, the game state engine of the particular game, as shown in FIG. **13**, may shift the virtual chip **1054**, **1056**, **1058**, or **1060** to the position mostly covered by the virtual chip **1054**, **1056**, **1058**, or **1060** to avoid any doubt of where the actual bet or wager resides. When doing so

the game state engine may also present a message on the display **1040** and/or generate an audible message to ensure that the bet or wager has been moved to the desired bet position.

(149) Once a player has concluded placing their baccarat bets or wagers, the player initiates the game by touching the deal button **1014** on display **1002** causing two virtual hands to be dealt, one for the bank and one for the player. In one embodiment, playing cards are animated to emulate physical playing cards being dealt. When dealt, the final value of the combination of cards for both the bank and the player are determined and are displayed in either the player display window **1068** or the banker display window **1070**. As dictated by the game rules, additional cards may be dealt to conclude a particular hand. Either based on the first two playing cards dealt to the banker and player or once any final playing cards are dealt to conclude a hand, the processor **1302**, shown in FIG. **13**, determines the final results and settles all bets or wagers. The final win settlement is shown in the win display **1052** and added to the current credits as shown in display **1046**.

(150) FIG. **11**A illustrates one embodiment of a user interface **1100** of a touch screen LCD button deck **1102** of the electronic table game of roulette according to the embodiments of the present invention. Although the user interface **1100** is utilized for the table game of roulette, it may be utilized for other table games where appropriate. A number of common elements may exist between various table game user interfaces, some elements may be common between such user interfaces. Such common elements may include a service button **1104** used to alert the casino staff there may be a problem with the EGM, a collect button **1106** which allows a player to collect or cash out their current credits generally by way of a printed credit voucher, a change game button **1108** which allows a player to change games by toggling the display to the select screen shown in FIG. **8** or similar, and a help/pays button **1110** which provides the player with game rules, game information, paytable and the like. For the particular game of roulette or similar games, a plurality of buttons are displayed allowing the player to clear a bet **1112**, undo a previous operation **1114**, or initiate the roulette game by touching the spin button **1116**.

(151) FIG. **11**B illustrates one embodiment of the top display **1120** of an EGM relating to the electronic table game of roulette according to the embodiments of the present invention. Those skilled in the art will recognize that any table game may be offered on a variety of display configurations including a single landscape mode display or a single portrait mode display with a separate button deck as illustrated in FIG. **11**A, a single landscape mode display or a single portrait mode display with button deck functionality included in the display, dual displays as illustrated herein, dual displays with the button deck functionality included in one of the two displays, or three displays, generally with landscape orientation, with either a separate button deck or button deck functionality included within one of the three displays.

(152) In the case of FIG. **11**B, one embodiment of the top display **1120** is comprised of either a touch screen display or non-touch screen display **1122**. Generally, the top display **1122** either represents an image of the table game being played, a logo of the game being played, a virtual dealer, or other images. As illustrated, the image displayed on top display **1122** is a virtual roulette table layout **1144** which includes a number of betting positions **1126**. Once the player places one or more bets or wagers, the player may initiate the games by touching the spin button **1116** as shown in FIG. **11**A causing the virtual roulette wheel **1124** to spin as the roulette ball also spins and/or bounces from one ending location **1128** to another ending location **1228**. As the virtual roulette wheel **1124** slows down, the roulette ball eventually comes to a final resting position in one of the virtual recesses **1128** which are associated with the final virtual wheel number **1126**. As shown, the roulette wheel **1124** includes 36 numbers along with zero and double zero positions. However, casino operators may configure the virtual roulette wheel **1124** to allow for a single zero position, the zero and double zero positions as shown, single, double, and triple zero positions, or single, double, triple and quadruple zero positions. As the number of zero positions changes, it has a dramatic effect on hold percentage of the game. For instance, a single zero roulette game has a hold percentage of 2.70%, a double zero wheel has a hold percentage of 5.26%, and a triple zero wheel

has a hold percentage of 7.69%. The roulette layout **1124** also includes a spin outcome history display **1132** which lists previous spin outcomes **1134**. The positioning of the outcomes on the left, center, or right may indicate red number on the left, green or zeros in the center and black numbers on the right. Those skilled in the art will recognize many differing positions may be implemented.

(153) In one embodiment, the roulette wheel spin is an animated sequence emulating real roulette wheels where the roulette wheel **1124** rotates with diminishing speed, eventually coming to a resting position while the roulette ball **1130** also spins and/or bounces around the virtual roulette wheel **1124** wherein the roulette ball **1130** eventually comes to a final resting position in one of the virtual recesses **1128** which are associated with the final virtual wheel number **1126**.

(154) FIG. **11**C illustrates one embodiment of the bottom display **1140** of an EGM relating to the electronic table game of roulette according to the embodiments of the present invention. In one embodiment, bottom display **1140** is comprised of a touch screen display **1142** displaying a roulette game layout **1144**. Bottom display **1140** also includes a number of generally common elements which may include a credit display **1146**, a bet or wager display **1148**, a menu button **1150** and a win display **1152**. Additionally, wager amounts are made by utilizing the virtual chips illustrated. The casino operator may determine and set bet or wager amounts as they desire. As an example, the interface illustrates a 250 chip **1154**, a $1 chip **1156**, a $5 chip **1158** and a $25 chip **1160**. However, a first casino operator may configure the betting range 50, 250, 500 and $1 while another operator may configure the betting range as $5, $25, $100, and $500. There are few limitations to how an operator may configure a betting range. Those skilled in the art will recognize that although four virtual chips of various denominations are illustrated, any number of virtual chips with varying denominations may be utilized. Bottom display **1140** also indicates the minimum bet or wager **1162** a player may make to initiate a game.

(155) In one embodiment, the roulette layout **1144** includes a number of betting or wagering positions **1164**. Generally, the game of roulette includes a great many betting or wagering positions, each offering a different type bet or wager with some offering the same payouts with some with differing payouts. The payouts generally escalate as the probability of winning decreases.

(156) Initially, when a new roulette game begins, a player places a bet or wager by using hand gestures between the virtual chips and the desired betting position **1164** on the roulette game layout **1144**. For instance, a player may touch a desired chip **1154**, **1156**, **1158**, or **1160** and slide it to the desired bet or wager position **1164** or touch a desired chip **1154**, **1156**, **1158**, or **1160**, lift the finger, and touch the desired betting or wagering position **1164**. Those skilled in the art will recognize that a variety of hand gestures may be utilized in this regard. As illustrated, a player has made a number of different bets or wagers as shown by virtual chips located on the roulette game layout **1144**. The bets or wagers shown include a $5 bet or wager **1166** on the first 12 bet, a $5 bet on the red color bet **1166**, $5 on the odd bet **1166** along with a number of $1 bets **1166** on the number bets. An operator may configure the game to allow for either a limited number of bets or wagers or an unlimited number of bets or wagers. This may be configured as the number of bets or wagers or by setting monetary minimum and maximum bets or wagers. Any number of bets or wagers may be placed on a betting or wagering position **1164**. As an example, a player may place a $1 virtual chip **1156** on red, then add two 250 virtual chip **1154** to the red bet, followed by a $25 virtual chip **1160** and two $5 virtual chip **1158** to the red bet for a total bet or wager of $36.50. As a player places their bets or wagers, the bet or wager amount is deducted from their credit balance as shown in credit display **1146**. The type and number of such bet or wager placements are generally only limited by the operator's desired configuration. In the event a player places the virtual chip **1154**, **1156**, **1158**, or **1160** on a betting or wagering position which is partially on one bet or wagering position and partially on another bet or wagering position, which may also occur in a physical game, the game state engine of the particular game, as shown in FIG. **13**, may shift the virtual chip **1154**, **1156**, **1158**, or **1160** to the position mostly covered by the virtual chip **1154**, **1156**, **1158**, or **1160** to avoid any doubt of where the actual bet or wager resides. However, in the game of roulette,

some wagers are intended to cover multiple bet positions and are thus accepted on such occasions.

(157) Once a player has concluded placing their bets or wagers, the player initiates the roulette game by touching the spin button **1116** on display **1100** causing the virtual roulette wheel **1124** to begin to spin. Following the virtual roulette wheel **1124** and roulette ball **1130** coming to rest, the processor **1302**, shown in FIG. **13**, determines the final results and settles all bets or wagers. The final win settlement is shown in the win display **1152** and then added to the current credits as shown in display **1146**.

(158) An important feature of EGMs is the number of plays possible within a given time frame, e.g., plays per hour. For some ETG games such as blackjack or roulette, there is slight lag between the time a player initiates the game and the conclusion of the game. However, for games which include virtual physical items to determine game outcomes such as the dice in the game of craps or the roulette wheel in the game of roulette, significant time is taken to allow for the animation of the dice coming to rest in the game of craps or the roulette wheel coming to rest and the roulette ball falling into a final position. Embodiments of the present invention allow for a player to touch either the roll button for craps or the spin button in roulette to immediately advance to the final outcome by truncating much or all of the game animation thereby speeding up the game or plays per hour by a considerable amount. Any embodiments of the present invention may similarly include such speed play functionality. In addition, any embodiments of the present invention may include play speed functionality which enables the player to adjust the speed of play of a particular game, i.e., the player can adjust the total time the animated dice roll in craps, adjust the total time a roulette wheel spins until the roulette ball comes to rest in the game of roulette, adjust the time required to deal a hand of blackjack, adjust the time required to deal baccarat, or similar. Play speed having a different functionality than speed play. The operator may also configure the game to automatically play without the animation thereby speeding up the game.

(159) FIG. **12**A illustrates one embodiment of a user interface **1200** of a touch screen LCD button deck **1202** of the electronic table game of blackjack according to the embodiments of the present invention. Although the user interface **1200** is utilized for the table game of blackjack, it may be utilized for other table games where appropriate. A number of common elements may exist between various table game user interfaces, some elements may be common between such user interfaces. Such common elements may include a service button **1204** used to alert the casino staff there may be a problem with the EGM, a collect button **1206** which allows a player to collect or cash out their current credits generally by way of a printed credit voucher, a change game button **1208** which allows a player to change games by toggling the display to the select screen shown in FIG. **8** or similar, and a help/pays button **1210** which provides the player with game rules, game information, paytable and the like. For the particular game of blackjack or similar games, a plurality of buttons are displayed which allow the player to clear a bet **1212**, receive another card via the hit button **1214**, double their bet button **1216**, stand button **1218**, and deal button **1219** to initiate the dealing of a blackjack hand. Buttons may also offer dual functionality such as changing to a split button in the event both of the player's cards are the same, e.g., two sevens, two nines, two queens, etc., regardless of suit.

(160) FIG. **12**B illustrates one embodiment of the top display **1220** of an EGM relating to the electronic table game ETG of blackjack according to the embodiments of the present invention. Those skilled in the art will recognize that any table game may be offered on a variety of display configurations including a single landscape mode display or a single portrait mode display with a separate button deck as illustrated in FIG. **12**A, a single landscape mode display or a single portrait mode display with button deck functionality included in the display, dual displays as illustrated herein, dual displays with the button deck functionality included in one of the two displays, or three displays, generally with landscape orientation, with either a separate button deck or button deck functionality included within one of the three display.

(161) In the case of FIG. **12**B, one embodiment of the top display **1220** is comprised of either a

touch screen display or non-touch screen display **1222**. Generally, the top display **1222** either represents an image of the table game being played, a logo of the game being played, a virtual dealer, or other images.

(162) FIG. **12**C illustrates one embodiment of the bottom display **1240** of an EGM relating to the electronic table game of blackjack according to the embodiments of the present invention. In one embodiment, bottom display **1240** is comprised of a touch screen display **1242** displaying a baccarat game layout **1244**. Bottom display **1240** also includes a number of generally common elements which may include a credit display **1246**, a bet or wager display **1248**, a menu button **1250** and a win display **1252**. Additionally, wager amounts are made by utilizing the virtual chips illustrated. The casino operator may determine and set bet or wager amounts as they desire. As an example, the interface illustrates a 250 chip **1254**, a $1 chip **1256**, a $5 chip **1258** and a $25 chip **1260**. However, a particular casino operator may configure the betting range 50, 250, 500 and $1 while another operator may configure the betting range as $5, $25, $100, and $500. There are few limitations to how an operator may configure a betting range. Those skilled in the art will recognize that although four virtual chips of various denominations are illustrated, any number of virtual chips with varying denominations may be utilized. Bottom display **1240** also indicates the minimum bet or wager **1262** a player may make to initiate a game.

(163) In one embodiment, the blackjack layout **1244** includes a number of betting or wagering positions player spot 1, player spot 2, or player spot 3 **1266**, each offering a different hand for the player. In one embodiment, the blackjack game includes a side bet known as a “Lucky Charlie” side bet **1268**. One skilled in the art will recognize many different type side bets may be added to the base blackjack game. In certain games, such as blackjack, side bets may be very important as they generally have a much higher hold percentage than the base game and generate a much higher win for the casino operator. For instance, the hold percentage of blackjack may be slightly above 1%, depending on the game rules and the player's skill. However, if the player also places a bet on the “Lucky Charlie” side bet **1268**, the hold percentage may increase to over 6%, depending on the amount of the “Lucky Charlie” side bet **1268**. Side bets often offer higher payout multiples to attract players as shown in paytable **1270**. Side bets may become of lesser importance on some table games as the games either have higher base hold percentages or offer various traditional bets with higher hold percentages, such as the game of craps.

(164) Initially, when a new blackjack game begins, a player places a bet or wager by using hand gestures between the virtual chips and the desired betting position **1272** on the blackjack layout **1244**. For instance, a player may touch a desired chip **1254**, **1256**, **1258**, or **1260** and slide it to the desired bet or wager position **1272** or touch a desired chip **1254**, **1256**, **1258**, or **1260**, lift the finger, and then touch the desired betting or wagering position **1272**. Those skilled in the art will recognize that a variety of hand gestures may be utilized in this regard. As illustrated, a particular player has made a number of different bets or wagers as shown by virtual chips located on the blackjack layout **1244**. The bets or wagers shown include $5 bets or wagers **1272** on player positions 1 and 2 and $1 bet or wager **1272** on the “Lucky Charlie” side bet **1268**. Generally, the game of blackjack only allows for a limited number of bets or wagers. Betting or wagering limitations may be configured as the number of bets or wagers or by setting monetary minimum and maximum bets or wagers. Any number of bets or wagers may be placed on a betting or wagering position **1266**. As a player places their bets or wagers, the bet or wager amount is deducted from their credit balance as shown in credit display **1248**. The type and number of such bet or wager placements are generally only limited by the operator's desired configuration. In the event a player places the virtual chip **1254**, **1256**, **1258**, or **1260** on a betting or wagering position which is partially on one bet or wagering position and partially on a non-betting or wagering position, which may also occur in a physical game, the game state engine of the particular game, as shown in FIG. **13**, may shift the virtual chip **1254**, **1256**, **1258**, or **1260** to the position closest to the virtual chip **1254**, **1256**, **1258**, or **1260** to avoid any doubt of where the actual bet or wager resides.

(165) Once a player has concluded placing their bets or wagers, the player initiates the blackjack game by touching the deal button **1219** on display **1202** causing one or more virtual hands to be dealt for the player and one for the dealer. In one embodiment, playing cards are animated to emulate playing cards being dealt in a real blackjack game and when dealt, one of the dealer's two cards are shown while the other card remains hidden (i.e., face down). The dealer's current hand total (excluding the face down card) may be displayed in dealer display window **1274**. As dictated by the game rules, additional cards may be dealt to the player upon request by the player touching the hit button **1214**, the double button **1216**, or the when both initial cards match, touching a split button (not shown) to divide the hand into two hands, both with the original bet or wager amount. In the alternative, the player may choose to stand by touching the stand button **1218** at any time. If the player's hand total exceeds 21, the hand ends immediately. If the player's first two cards are any ace with any card with a value of 10, the hand may end with an immediate payout of 3 to 2 or if enabled, the game may ask the player for “insurance” if the dealer's face up card is also an ace. Although the blackjack payout is listed as 3 to 2, the game is configurable to provide other payouts such as 6 to 5 or even money, as an example. Once a player has concluded their play without “busting” or exceeding a total hand value of 21 or less, the player may touch the stand button **1218**. If the player has placed a bet or wager on additional gaming positions 2 or 3 to form two or three total hands, the player sequentially concludes each additional hand by either “busting” or touching the stand button **1218**. If one or more of the player's hands are still in play without “busting,” the dealer plays out their hand. The dealer's options are limited as the dealer is required to “draw to 16 and stand on 17 or above.” In the event the dealer receives an ace as their up card, the game may either offer insurance or reveal a blackjack, if one has occurred. Once a hand is concluded, the processor **1302**, shown in FIG. **13**, determines the final results and settles all bets or wagers. The final win settlement is shown in the win display **1252** and then added to the current credits as shown in display **1246**.
(166) Those skilled in the art will recognize that although the embodiments illustrated in FIGS. **9** through **12** are shown with dual landscape-oriented displays, they may be combined into a single portrait-oriented display.
(167) FIG. **13** illustrates a block diagram **1300** depicting discrete game engines for a plurality of electronic table games according to the embodiments of the present invention which includes a processor **1302**. Most game processing on EGMs includes one and only one game state engine as that is all that is required for game operation. This includes a great many multi-play EGMs games. However, due to the many complexities of table games, the vast differences in rules and game play, and the potentially many configurability options, embodiments of the present invention allow for separate and discrete game engines for each game or group of games.
(168) Initially, a player is presented with a game select screen similar to that illustrated in FIG. **8**. This step is shown in FIG. **13** as the game select step **1304**. Once the game is selected, the processor **1302** determines which game state engine to utilize generally corresponding to the particular game the player has selected. Although FIG. **13** illustrates the four games of craps, baccarat, roulette and blackjack, any number or types of table games may be included. For instance, the games depicted on the display may include a plurality of table games such as Craps, Baccarat, Roulette, Blackjack, Three-Card Poker, Pai Gow Poker, Caribbean Stud, Big 6, Keno, Bingo, Super Fun 21, Royal Match Blackjack, and the like.
(169) As illustrated in FIG. **13**, once a player has selected their game, the processor **1302** will route the play to the appropriate game state engine. The game state engines may also include or communicate with a database maintaining math models for each game and any side bets associated therewith. In the event the game selected is craps, the processor routes the play to the craps game state engine **1310** which processes the play in discrete steps with the first state being the bet state **1312**. Once the bet state is concluded, the roll state **1314** processes the rolling of the dice. Once concluded, the processor **1302** enters the evaluation state **1316** and once concluded moves to the

primary results state **1318** and finally to the final results state **1320**.
(170) In the event the game selected is baccarat, the processor **1302** routes the play to the baccarat game state engine **1330** which processes the play in discrete steps with the first state being the bet state **1332**. Once the bet state is concluded, the deal state **1334** processes the dealing of playing cards. Once concluded, the processor **1302** enters the evaluation state **1336** and once concluded, moves to the final results state/settle wager step **1338** and play is concluded.
(171) In the event the game selected is roulette, the processor **1302** routes the play to the roulette game state engine **1350** which processes the play in discrete steps with the first state being the bet state **1352**. Once the bet state is concluded, the wheel state **1354** processes the roulette wheel operation. Once concluded, the processor **1302** enters the evaluation state **1356** and once concluded moves to the final results state/settle wager step **1358** and play is concluded.
(172) In the event the game selected is blackjack, the processor **1302** routes the play to the blackjack game state engine **1370** which processes the play in discrete steps with the first state being the bet state **1372**. Once the bet state is concluded, the deal state **1374** processes the dealing operation. Once concluded, the processor **1302** enters the player action state **1376** which determines what player actions have occurred and then enters the dealer action state **1378**. Once the player action state **1376** and dealer action state **1378** have concluded, processing enters the evaluation state **1380** and once concluded moves to the final results state/settle wager step **1382** and play is concluded.
(173) FIG. **14** illustrates an expanded block diagram of the craps game engine **1310** of FIG. **13** for electronic table games according to the embodiments of the present invention. The craps state engine **1310** is included within the complete craps processing **1400**. Once a player has selected the game of craps from the plurality of games offered, the game enters the bet state **1312** wherein the processor determines if the player has enough credits **1402** to meet the game's minimum bet or wager configuration. In the event credits are insufficient to meet the minimum bet or wager, the player is alerted **1404** to add credits to meet the minimum bet or wager configuration. If the player has sufficient credits to place a bet or wager, the player may continue the game and place a bet or wager. If the player does not place a bet or wager, as determined in decision block **1406**, the player is alerted to place a bet or wager **1408**. If the player has placed a bet or wager the processor determines if the bet or wager is within the minimum and maximum limits **1410**. If the bet or wager is outside of the minimum and maximum limits, the player is notified to either add additional bet or wager or they have reached the games maximum limit **1412**. If the bet or wager is within minimum and maximum limits **1410**, the player is allowed to press the play button to start the game.
(174) Once the player presses the play button, the game processing enters the bet state **1314** and begins with the roll of the virtual or graphical dice. If the player has pressed the speed play button **1422**, the virtual dice immediately come to rest at **1424**. If not, the virtual dice and animated and simulate a dice roll and eventually come to a final resting position **1426**. Once the simulated virtual dice roll comes to a final resting position, the processor instructs the game to go to the evaluation state **1316**. Once entering the evaluation state **1316**, the processor loops through each recorded bet or wager to resolve wins and losses **1430**.
(175) Once wins and losses are resolved **1430** the game enters the primary results state **1318**. Once in the primary results state **1318** resolves bets or wagers, winning bets or wagers are paid or losing bets or wagers removed **1432**. Once resolved, check win against lockup amount limit occurs **1434**. Following this occurrence, a determination is made if there are any bets unresolved **1436**. If no unresolved bets remain, the player is either paid for winning bets or wagers or losing bets or wagers are removed **1438**. If unresolved bets or wagers remain, a point is set wherein the game continues to the next roll **1442**. A similar scenario or sequence of events occurs for come line bets.
(176) Once primary results state **1318** concludes, the processor enters the final results state **1320**. Initially, the processor checks win against lockup amount limit **1444**. Thereafter, the game history

is posted to the platform **1446** as is required in most gaming jurisdictions. Not every game ever played is required to reside in the game history. Instead, only a predetermined number of games or actions need be posted for a later recall to check the game history. As a limited number of games or actions need be recorded, a game history limit is established, e.g., last 25 games, last 50 games, last 100 games, etc. A determination is made if the current game history, when added to the previously posted games exceed the history limits **1448**. If not, the game history is recorded and posted **1450**. If the game history exceeds the limits, the oldest posted game or actions are deleted and the current posted game is added to the history as the most recent game or action **1452**. At this point, the processor causes the entire process **1400** to begin again for the next game to be played.

(177) FIG. **15** illustrates an expanded block diagram **1500** of the baccarat game engine of FIG. **13** for electronic table games according to the of the present invention. The baccarat state engine **1330** is included within the complete baccarat processing **1500**. Once a player has selected the game of baccarat from the plurality of games offered, the game enters the bet state **1332** wherein the processor determines if the player has enough credits **1502** to meet the game's minimum bet or wager configuration. In the event credits are insufficient to meet the minimum bet or wager, the player is alerted **1504** to add credits to meet the minimum bet or wager configuration. If the player has sufficient credits to place a bet or wager, the player may continue the game and place a bet or wager. If the player does not place a bet or wager, as determined in decision block **1506**, the player is alerted to place a bet or wager **1508**. If the player has placed a bet or wager the processor determines if the bet or wager is within the minimum and maximum limits **1510**. If the bet or wager is outside of the minimum and maximum limits, the player is notified to either add additional bet or wager or they have reached the games maximum limit **1512**. If the bet or wager is within minimum and maximum limits **1510**, the player is allowed to press the play button to start the game **1514**.

(178) Once the player presses the play button, the game processing then enters the deal state **1334** and player and banker cards are dealt **1530**. The game then enters the evaluation state **1336**, the processor loops through each recorded bet or wager to resolve wins and losses **1532**.

(179) Once evaluation state **1336** concludes, the processor enters the final results state **1338**. Initially, the processor checks win against lockup amount limit **1544**. Thereafter, the game history is posted to the platform **1546** as is required in most gaming jurisdictions. Not every game ever played is required to reside in the game history. Instead, only a predetermined number of games or actions need be posted for a later recall to check the game history. As a limited number of games or actions need be recorded, a game history limit is established, e.g., last 25 games, last 50 games, last 100 games, etc. A determination is made if the current game history, when added to the previously posted games exceed the history limits **1548**. If not, the game history is recorded and posted **1550**. If the game history exceeds the limits, the oldest posted game or actions are deleted and the current posted game is added to the history as the most recent game or action **1552**. At this point, the processor causes the entire process **1500** to begin again for the next game to be played.

(180) FIG. **16** illustrates an expanded block diagram **1600** of the roulette game engine of FIG. **13** for electronic table games according to the embodiments of the present invention. The roulette state engine **1350** is included within the complete roulette processing. Once a player has selected the game of roulette from the plurality of games offered, the game enters the bet state **1352** wherein the processor determines if the player has enough credits **1602** to meet the game's minimum bet or wager configuration. In the event credits are insufficient to meet the minimum bet or wager, the player is alerted **1604** to add credits to meet the minimum bet or wager configuration. If the player has sufficient credits to place a bet or wager, the player may continue the game and place a bet or wager. If the player does not place a bet or wager, as determined in decision block **1606**, the player is alerted to place a bet or wager **1608**. If the player has placed a bet or wager the processor determines if the bet or wager is within the minimum and maximum limits **1610**. If the bet or wager is outside of the minimum and maximum limits, the player is notified to either add

additional bet or wager or they have reached the games maximum limit **1612**. If the bet or wager is within minimum and maximum limits **1610**, the player is allowed to press the play button to start the game.

(181) Once the player presses the play button, the game processing then enters the wheel state **1354** and begins with the spin the virtual roulette wheel and roulette ball. If the player has pressed the speed play button **1622**, the virtual dice immediately come to rest at **1626**. If not, the virtual dice simulate a dice roll and eventually come to a final resting position **1624**. Once the simulated graphical dice roll comes to a final resting position, the processor instructs the game to go to the evaluation state **1628** and then enters the evaluation state **1356**. Once entering the evaluation state **1356**, the processor loops through each recorded bet or wager to resolve wins and losses **1630**.

(182) Once evaluation state **1356** concludes, the processor enters the final results state **1358**. Initially, the processor checks win against lockup amount limit **1632**. Thereafter, the game history is posted to the platform **1634** as is required in most gaming jurisdictions. Not every game ever played is required to reside in the game history. Instead, only a predetermined number of games or actions need be posted for a later recall to check the game history. As a limited number of games or actions need be recorded, a game history limit is established, e.g., last 25 games, last 50 games, last 100 games, etc. A determination is made if the current game history, when added to the previously posted games exceed the history limits **1636**. If not, the game history is recorded and posted **1638**. If the game history exceeds the limits, the oldest posted game or actions are deleted and the current posted game is added to the history as the most recent game or action **1640**. At this point, the processor causes the entire process **1400** to begin again for the next game to be played.

(183) FIG. **17** illustrates an expanded block diagram **1700** of the blackjack game engine of FIG. **13** for electronic table games according to the embodiments of the present invention. The blackjack state engine **1370** is included within the complete blackjack processing. Once a player has selected the game of blackjack from the plurality of games offered, the game enters the bet state **1372** wherein the processor determines if the player has enough credits **1702** to meet the game's minimum bet or wager configuration. In the event credits are insufficient to meet the minimum bet or wager, the player is alerted **1704** to add credits to meet the minimum bet or wager configuration. If the player has sufficient credits to place a bet or wager, the player may continue the game and place a bet or wager. If the player does not place a bet or wager, as determined in decision block **1706**, the player is alerted to place a bet or wager **1708**. If the player has placed a bet or wager the processor determines if the bet or wager is within the minimum and maximum limits **1710**. If the bet or wager is outside of the minimum and maximum limits, the player is notified to either add additional bet or wager or they have reached the games maximum limit **1712**. If the bet or wager is within minimum and maximum limits **1710**, the player is allowed to press the play button to start the game **1714**.

(184) Once the player presses the play button, the game processing then enters the deal state **1374** and player and dealer cards are dealt **1730**. Once the initial player and dealer cards are dealt, the game enters the player action state **1376** wherein the processor determines if the player has an adequate credit balance in the event the player doubles or splits their cards per game rules **1732** after which the player is allowed to continue play and either hit or receives one or more additional cards, stands on the two cards originally dealt, doubles their bet or wager per game rules or splits their original two cards if they match in value per game rules and play two separate hands as opposed to a single hand.

(185) Once the player action state **1376** concludes, the processor enters the dealer action state **1378** wherein a determination is made as to whether the dealer's first two cards equal twenty-one **1736**, known as a blackjack. If the dealer has a blackjack, the processor immediately reveals the dealer's hole card **1738** and the game concludes for any hand where the player does not also have a blackjack where the hand is considered a tie. If the dealer does not have a blackjack on their original two cards, play continues and the dealer draws cards according to the rules **1740**. The

game then enters the evaluation state **1380**, the processor loops through each recorded bet or wager to resolve wins and losses **1742**.

(186) Once evaluation state **1380** concludes, the processor enters the final results state **1382**. Initially, the processor checks win against lockup amount limit **1744**. Thereafter, the game history is posted to the platform **1746** as is required in most gaming jurisdictions. Not every game ever played is required to reside in the game history. Instead, only a predetermined number of games or actions need be posted for a later recall to check the game history. As a limited number of games or actions need be recorded, a game history limit is established, e.g., last 25 games, last 50 games, last 100 games, etc. A determination is made if the current game history, when added to the previously posted games exceeds the history limits **1458**. If not, the game history is recorded and posted **1750**. If the game history exceeds the limits, the oldest posted game or actions are deleted and the current posted game is added to the history as the most recent game or action **1752**. At this point, the processor causes the entire process **1700** to begin again for the next game to be played. Those skilled in the art will recognize that the same or similar sequence of events occur if the players has placed multiple bets or wagers. Although the embodiments of the present invention have been described in detail in the block diagrams of FIGS. **14** through **17**, with reference to several embodiments, additional variations and modifications exist within the scope and spirit of the invention. Moreover, other than those illustrated, more or less steps or actions may be included or the sequencing may change within the scope and spirit of the invention.

(187) FIG. **18** illustrates a combined single display of the embodiment illustrated in FIG. **9** relating to the table game of craps according to the embodiments of the present invention. All functionality, with the exception of the button panel and animated dice, as described in FIG. **9**A through FIG. **9**C, remains the same and need not be repeated here.

(188) As illustrated, the button panel of FIG. **9**A is incorporated into the embodiment illustrated in FIG. **18** by a combination of physical and programmable LCD buttons with only one game display. This display type is particularly well suited for single display EGMs such as bar-top EGMs or slant type EGMs. Those skilled in the art will recognize that any combination of physical or programmable LCD buttons may be utilized, including all physical buttons or all LCD buttons.

(189) Although the button deck **1802** is utilized for the table game of craps, it may be utilized for other table games where appropriate. A number of common elements may exist between various table game user interfaces, some elements may be common between such user interfaces. Such common elements may include a service button **1810** used to alert the casino staff there may be a problem with the EGM, a collect button **1812** which allows a player to collect or cash out their current credits generally by way of a printed credit voucher, a change game button **1814** which allows a player to change games by toggling the display to the select screen shown in FIG. **8** or similar, and a help/pays button **1816** which provides the player with game rules, game information, paytable and the like. For the particular game of craps or similar games, a plurality of programmable LCD buttons are displayed which allow the player to clear a bet **1818**, undo a previous operation **1822**, or initiate the craps game by touching the roll button **1826**. As illustrated, the programmable LCD buttons **1820** and **1824** are programmed to be blank and not used for the craps embodiment of the present invention. Alternatives, in any embodiment, include dimming the buttons, using different coloring, listing as “not used,” etc.

(190) After a player places their bets or wagers, the player touches the roll button **1826** to begin the roll portion of the craps game **940**′. In the case of a single display as illustrated, the virtual dice **928**′ appear to roll on the craps game layout **941**′ as opposed to the upper display **920** as illustrated in FIG. **9**B and the game is concluded with bets or wagers settled.

(191) FIG. **19** illustrates a combined single display of the embodiment illustrated in FIG. **10** relating to the table game of baccarat according to the embodiments of the present invention. All functionality, with the exception of the button panel, as described in FIG. **10**A through FIG. **10**C, remains the same and is not repeated here.

(192) As illustrated, the button panel of FIG. **10**A is incorporated into the embodiment illustrated in FIG. **19** by a combination of physical and programmable LCD buttons with only one game display. This display type is particularly well suited for single display EGMs such as bar-top EGMs or slant type EGMs. Those skilled in the art will recognize that any combination of physical or programmable LCD buttons may be utilized, including all physical buttons or all LCD buttons.
(193) Although the button deck **1802** is utilized for the table game of baccarat, it may be utilized for other table games where appropriate. A number of common elements may exist between various table game user interfaces, some elements may be common between such user interfaces. Such common elements may include a service button **1810** used to alert the casino staff there may be a problem with the EGM, a collect button **1812** which allows a player to collect or cash out their current credits generally by way of a printed credit voucher, a change game button **1814** which allows a player to change games by toggling the display to the select screen shown in FIG. **8** or similar, and a help/pays button **1816** which provides the player with game rules, game information, paytable and the like. For the particular game of baccarat or similar games, a plurality of programmable LCD buttons are displayed which allow the player to clear a bet **1820** or initiate the craps game by touching the deal button **1826**. As illustrated, the programmable LCD buttons **1818**, **1822**, and **1824** are programmed to be blank and not used for the baccarat embodiment of the present invention.
(194) After a player places their bets or wagers on layout **1044′**, the player touches the deal button **1826** to begin the deal portion of the baccarat game **1040′** and conclude the game with bets or wagers settled.
(195) FIG. **20** illustrates a combined single display of the embodiment illustrated in FIG. **11** relating to the table game of roulette according to the embodiments of the present invention. All functionality, with the exception of the button panel and animated wheel and roulette ball, as described in FIGS. **11**A through **11**C, remains the same and need not be repeated here.
(196) As illustrated, the button panel of FIG. **11**A is incorporated into the embodiment illustrated in FIG. **20** by a combination of physical and programmable LCD buttons with only one game display. This display type is particularly well suited for single display EGMs such as bar-top EGMs or slant type EGMs. Those skilled in the art will recognize that any combination of physical or programmable LCD buttons may be utilized, including all physical buttons or all LCD buttons.
(197) Although the button deck **1802** is utilized for the table game of roulette, it may be utilized for other table games where appropriate. A number of common elements may exist between various table game user interfaces, some elements may be common between such user interfaces. Such common elements may include a service button **1810** used to alert the casino staff there may be a problem with the EGM, a collect button **1812** which allows a player to collect or cash out their current credits generally by way of a printed credit voucher, a change game button **1814** which allows a player to change games by toggling the display to the select screen shown in FIG. **8** or similar, and a help/pays button **1816** which provides the player with game rules, game information, paytable and the like. For the particular game of roulette or similar games, a plurality of programmable LCD buttons are displayed which allows the player to clear a bet **1818**, undo a previous operation **1822**, or initiate the roulette game by touching the spin button **1826**. As illustrated, the programmable LCD buttons **1820** and **1824** are programmed to be blank and not used for the roulette embodiment of the present invention.
(198) After a player places their bets or wagers on layout **1144′**, the player touches the spin button **1826** to begin the spin portion of the roulette game **1140′**. In the case of a single display as illustrated, the virtual wheel **1124′** rotates or spins along with the spinning or bouncing of roulette ball **1130′** until the virtual wheel **1124′** comes to rest along with roulette ball **1130′**. The game is concluded with bets or wagers settled. A history display **1132′** maintains a record of the most recent game outcomes.
(199) FIG. **21** illustrates a combined single display of the embodiment illustrated in FIG. **12**

relating to the table game of blackjack according to the embodiments of the present invention. All functionality, with the exception of the button panel, as described in FIGS. **12**A through **12**C, remains the same and need not be repeated here.

(200) As illustrated, the button panel of FIG. **12**A is incorporated into the embodiment illustrated in FIG. **21** by a combination of physical and programmable LCD buttons with only one game display. This display type is particularly well suited for single display EGMs such as bar-top EGMs or slant type EGMs. Those skilled in the art will recognize that any combination of physical or programmable LCD buttons may be utilized, including all physical buttons or all LCD buttons.

(201) Although the button deck **1802** is utilized for the table game of blackjack, it may be utilized for other table games where appropriate. A number of common elements may exist between various table game user interfaces, some elements may be common between such user interfaces. Such common elements may include a service button **1810** used to alert the casino staff there may be a problem with the EGM, a collect button **1812** which allows a player to collect or cash out their current credits generally by way of a printed credit voucher, a change game button **1814** which allows a player to change games by toggling the display to the select screen shown in FIG. **8** or similar, and a help/pays button **1816** which provides the player with game rules, game information, paytable and the like. For the particular game of blackjack or similar games, a plurality of programmable LCD buttons are displayed which allow the player to clear a bet **1818**, ask for an additional card or hit via button **1820**, double their bet button **1822**, stand via button **1824** or start or initiate the blackjack game by touching the deal button **1826**. As illustrated, in this case and similar, there is no need for any of the programmable LCD buttons to be left blank. In the event the player receives identical value cards on the initial deal, one of the programmable LCD buttons displays a split message wherein the players may divide the hand into two separate hands and double their bet. Those skilled in the art will recognize that in any blackjack or similar game embodiment where additional hands are derived from an original hand, the player may be offered the choice of side bets related to the secondary hands.

(202) After a player places his or her blackjack bets or wagers on layout **1244′**, the player touches the deal button **1826** to begin the deal portion of the blackjack game **1240′** and after the player's and dealer's hands are concluded the game, the bets or wagers are settled.

(203) In one embodiment, the EGM only offers table games. That is, the EGM does not offer non-table games such as video poker, keno, bingo, etc. In this manner, the EGM provides a teaching tool for players to experience how table games operate before venturing onto a physical, and possibly intimidating, casino table game. In one embodiment, the table game EGM may be positioned near the casino pit which accommodates the physical casino table games.

(204) Although the invention has been described in detail with reference to several embodiments, additional variations and modifications exist within the scope and spirit of the invention as described and defined in the following claims.

Claims

1. A gaming system comprising: a monetary input device configured to receive a physical item associated with a monetary value; a processor running executable instructions and associated memory; one or more displays with at least a primary display area for playing virtual casino table games; one or more user interfaces configured, along with the processor and associated memory, to allow a player to at least select a virtual casino table game to play from two or more stored virtual table games, place bets and start a hand or round of the selected virtual casino table game; a plurality of game state engines corresponding to at least two or more stored virtual casino table games including at least two table games from a group of table games consisting of craps, baccarat, roulette, blackjack, three-card poker, pai gow poker, Caribbean stud and big 6 wheel, wherein a sequence of play for each virtual casino table game is controlled by a dedicated one of the plurality

of game state engines; wherein the processor is configured to allow the player to adjust speed of play associated with the virtual casino table games and wherein the processor truncates game animation associated with stored virtual table games when played to adjust the play speed; wherein for at least the games of roulette and craps, an upper display is used to depict at least a portion of game play while a lower display area is used to depict at least a portion of a digital betting game layout for at least one of the table games; and wherein at least one side bet is offered to the player with at least one side bet having a higher hold percentage than the associated virtual casino table game.

2. The gaming system of claim 1 wherein at least one of the at least one side bets increases the average hold percentage of the associated virtual casino table game when played.

3. The gaming system of claim 1 wherein at least one of the at least one side bets has a payout which is a multiple of the associated virtual casino table game side bet wager.

4. The gaming system of claim 1 wherein at least one of the at least one side bets has a payout which is based on a progressive value of the associated virtual casino table game side bet.

5. The gaming system of claim 4 wherein the seed amount of the progressive value is greater than the side bet wager amount.

6. The gaming system of claim 1 wherein at least one of the plurality of side bets increases the average overall hold percentage of the gaming system when played.

7. The gaming system of claim 6 wherein the increase of the average overall hold percentage is based at least in part on a progressive virtual casino table game side bet when played.

8. The gaming system of claim 1 further comprising two user interfaces in the form of a touch screen primary display and a button deck comprising a programmable LCD screen.

9. The gaming system of claim 8 wherein the programmable LCD screen is configured to depict a different arrangement of potential inputs corresponding to each of the two or more stored virtual casino table games.

10. The gaming system of claim 1 further comprising a selectable chip denomination module whereby an operator may select denominations of virtual betting chips depicted on the primary display.

11. The gaming system of claim 1 wherein only virtual casino table games are stored.

12. The gaming system of claim 1 wherein each game state engine includes a math model associated with the corresponding stored virtual table game.

13. The gaming system of claim 1 wherein an upper display area and a lower display area are located on two or more displays.

14. The gaming system of claim 13 wherein an upper display area and a lower display area are located on two or more displays with the displays positioned in a landscape orientation.

15. The gaming system of claim 1 wherein an upper display area and a lower display area are located on a single display.

16. The gaming system of claim 15 wherein the single display is positioned in a portrait orientation.

17. A gaming system comprising: a monetary input device configured to receive a physical item associated with a monetary value; a processor and associated memory; one or more displays with at least a primary display area for playing virtual casino table games; one or more user interfaces configured, along with the processor and associated memory, to allow a player to at least select a virtual casino table game to play from two or more stored virtual table games, place bets and start a hand or round of the selected virtual casino table game; a plurality of game state engines corresponding to at least two or more stored virtual casino table games including at least two table games from a group of table games consisting of craps, baccarat, roulette, blackjack, three-card poker, pai gow poker, Caribbean stud and big 6 wheel, wherein a sequence of play for each virtual casino table game is controlled by a dedicated one of the plurality of game state engines; wherein the processor is configured to allow the player to adjust speed of play associated with the virtual casino table games; wherein an upper display area is used to depict at least a portion of game play

while a lower display area is used to depict at least a portion of a digital betting game layout for at least one of the table games; and wherein at least side bet is offered to the player with at least one side bet having a higher hold percentage than the associated virtual casino table game.

18. The gaming system of claim 17 wherein at least one of the at least one side bets increases the average hold percentage of the associated virtual casino table game when played.

19. The gaming system of claim 17 wherein at least one of the at least one side bets has a payout which is a multiple of the associated virtual casino table game side bet wager.

20. The gaming system of claim 17 wherein at least one of the at least one side bets has a payout which is based on a progressive value of the associated virtual casino table game side bet.

21. The gaming system of claim 20 wherein the seed amount of the progressive value is greater than the side bet wager amount.

22. The gaming system of claim 17 wherein the at least one side bet offered to the player increases the average overall hold percentage of the gaming system when played.

23. The gaming system of claim 22 wherein the increase of the average overall hold percentage is based at least in part on a progressive virtual casino table game side bet when played.

24. The gaming system of claim 17 further comprising two user interfaces in the form of a touch screen primary display and a button deck comprising a programmable LCD screen.

25. The gaming system of claim 24 wherein the programmable LCD screen is configured to depict a different arrangement of potential inputs corresponding to each of the two or more stored virtual casino table games.

26. The gaming system of claim 17 further comprising a selectable chip denomination module whereby an operator may select denominations of virtual betting chips depicted on the primary display.

27. The gaming system of claim 17 wherein only virtual casino table games are stored.

28. The gaming system of claim 17 wherein each game state engine includes a math model associated with the corresponding stored virtual table game.

29. The gaming system of claim 17 wherein an upper display area and a lower display area are located on two or more displays.

30. The gaming system of claim 29 wherein an upper display area and a lower display area are located on two or more displays with the displays positioned in a landscape orientation.

31. The gaming system of claim 17 wherein an upper display area and a lower display area are located on a single display.

32. The gaming system of claim 31 wherein the single display is positioned in a portrait orientation.

33. A gaming system comprising: a monetary input device configured to receive a physical item associated with a monetary value; a processor and associated memory; one or more displays with at least a primary display area for playing virtual casino table games; one or more user interfaces configured, along with the processor and associated memory, to allow a player to at least select a virtual casino table game to play from two or more stored virtual table games, place bets and start a hand or round of the selected virtual casino table game; a plurality of game state engines corresponding to at least two or more stored virtual casino table games including at least two table games from a group of table games consisting of craps, baccarat, roulette, blackjack, three-card poker, pai gow poker, Caribbean stud and big 6 wheel, wherein a sequence of play for each virtual casino table game is controlled by a dedicated one of the plurality of game state engines; wherein an upper display area is used to depict at least a portion of game play while a lower display area is used to depict at least a portion of a digital betting game layout for at least one of the table games; and wherein at least one side bet is offered to the player with at least one side bet having a higher hold percentage than the associated virtual casino table game.

34. The gaming system of claim 33 wherein at least one of the at least one side bets increases the average hold percentage of the associated virtual casino table game when played.

35. The gaming system of claim 33 wherein at least one of the at least one side bets has a payout

which is a multiple of the associated virtual casino table game side bet wager.
36. The gaming system of claim 33 wherein at least one of the at least one side bets has a payout which is based on a progressive value of the associated virtual casino table game side bet.
37. The gaming system of claim 36 wherein the seed amount of the progressive value is greater than the side bet wager amount.
38. The gaming system of claim 33 wherein the at least one side bet increases the average overall hold percentage of the gaming system when played.
39. The gaming system of claim 38 wherein the increase of the average overall hold percentage is based at least in part on a progressive virtual casino table game side bet when played.
40. The gaming system of claim 33 further comprising two user interfaces in the form of a touch screen primary display and a button deck comprising a programmable LCD screen.
41. The gaming system of claim 40 wherein the programmable LCD screen is configured to depict a different arrangement of potential inputs corresponding to each of the two or more stored virtual casino table games.
42. The gaming system of claim 33 further comprising a selectable chip denomination module whereby an operator may select denominations of virtual betting chips depicted on the primary display.
43. The gaming system of claim 33 wherein only virtual casino table games are stored.
44. The gaming system of claim 33 wherein each game state engine includes a math model associated with the corresponding stored virtual table game.
45. The gaming system of claim 33 wherein an upper display area and a lower display area are located on two or more displays.
46. The gaming system of claim 45 wherein an upper display area and a lower display area are located on two or more displays with the displays positioned in a landscape orientation.
47. The gaming system of claim 33 wherein an upper display area and a lower display area are located on a single display.
48. The gaming system of claim 47 wherein the single display is positioned in a portrait orientation.
49. A gaming method comprising: configuring a monetary input device to receive a physical item associated with a monetary value; configuring at least one processor running executable instructions to operate a game of chance on a gaming machine including one or more displays and memory in communication with the at least one processor, the one or more displays having at least a primary display area for playing virtual casino table games; configuring one or more user interfaces, along with the processor and associated memory, to allow a player to at least select a virtual casino table game to play from two or more stored virtual table games, place bets and start a hand or round of the selected virtual casino table game; further configuring the processor running executable instructions to utilize a plurality of game state engines corresponding to at least two or more stored virtual casino table games including at least two table games from a group of table games consisting of craps, baccarat, roulette, blackjack, three-card poker, pai gow poker, Caribbean stud and big 6 wheel, wherein a sequence of play for each virtual casino table game is controlled by a dedicated one of the plurality of game state engines; further configuring the processor running executable instructions to allow the player to adjust speed of play associated with the virtual casino table games and wherein the processor truncates game animation associated with stored virtual table games when played to adjust the play speed; utilizing an upper display area to depict at least a portion of game play while a lower display area is used to depict at least a portion of a digital betting game layout for at least one of the table games; offering at least one side bet to the player with at least one side bet having a higher hold percentage than the associated virtual casino table game; and enabling the player to select a wager for a game of chance and enabling the player to initiate a cash out operation.
50. The gaming method of claim 49 wherein at least one of the at least one side bets increases the average hold percentage of the associated virtual casino table game when played.

51. The gaming method of claim 49 wherein at least one of the at least one side bets has a payout which is a multiple of the associated virtual casino table game side bet wager.
52. The gaming method of claim 49 wherein at least one of the at least one side bets is based on a progressive value of the associated virtual casino table game side bet.
53. The gaming method of claim 52 wherein the seed amount of the progressive value is greater than the side bet wager amount.
54. The gaming method of claim 49 wherein at least one side bet offered to the player increases the average overall hold percentage of the gaming system when played.
55. The gaming method of claim 54 wherein the increase of the average overall hold percentage is based at least in part on a progressive virtual casino table game side bet when played.
56. The gaming method of claim 49 further comprising two user interfaces in the form of a touch screen primary display and a button deck comprising a programmable LCD screen.
57. The gaming method of claim 56 wherein the programmable LCD screen is configured to depict a different arrangement of potential inputs corresponding to each of the two or more stored virtual casino table games.
58. The gaming method of claim 49 further comprising a selectable chip denomination module whereby an operator may select denominations of virtual betting chips depicted on the primary display.
59. The gaming method of claim 49 wherein only virtual casino table games are stored.
60. The gaming method of claim 49 wherein each game state engine amongst the plurality of game state engines includes a math model associated with the corresponding stored virtual table game.
61. The gaming method of claim 49 wherein an upper display area and a lower display area are located on two or more displays.
62. The gaming method of claim 61 wherein an upper display area and a lower display area are located on two or more displays with the displays positioned in a landscape orientation.
63. The gaming method of claim 49 wherein an upper display area and a lower display area are located on a single display.
64. The gaming method of claim 63 wherein the single display is positioned in a portrait orientation.
65. A gaming method comprising: utilizing a monetary input device configured to receive a physical item associated with a monetary value; enabling a player to select a wager for a game of chance and enabling the player to initiate a cash out operation; configuring at least one processor running executable instructions to operate a game of chance on a gaming machine including one or more displays and memory in communication with the at least one processor, the one or more displays having at least a primary display area for playing virtual casino table games; configuring one or more user interfaces, along with the processor and associated memory, to allow a player to at least select a virtual casino table game to play from two or more stored virtual table games, place bets and start a hand or round of the selected virtual casino table game; configuring the processor running executable instructions to utilize a plurality of game state engines corresponding to at least two or more stored virtual casino table games including at least two table games from a group of table games consisting of craps, baccarat, roulette, blackjack, three-card poker, pai gow poker, Caribbean stud and big 6 wheel, wherein a sequence of play for each virtual casino table game is controlled by a dedicated one of the plurality of game state engines; further configuring the processor running executable instructions to allow the player to adjust speed of play associated with the virtual casino table games; utilizing an upper display area to depict at least a portion of game play while a lower display area is used to depict at least a portion of a digital betting game layout for at least one of the table games; and offering at least one side bet to the player with at least one side bet having a higher hold percentage than the associated virtual casino table game.
66. The gaming method of claim 65 wherein at least one of the at least one side bets increases the average hold percentage of the associated virtual casino table game when played.

67. The gaming method of claim 65 wherein at least one of the at least one side bets has a payout which is a multiple of the associated virtual casino table game side bet wager.
68. The gaming method of claim 65 wherein at least one of the at least one side bets has a payout which is based on a progressive value of the virtual casino table game side bet.
69. The gaming method of claim 68 wherein the seed amount of the progressive value is greater than the side bet wager amount.
70. The gaming method of claim 65 wherein the at least one side bet increases the average overall hold percentage of the gaming system when played.
71. The gaming method of claim 70 wherein the increase of the average overall hold percentage is based at least in part on a progressive virtual casino table game side bet when played.
72. The gaming method of claim 65 further comprising two user interfaces in the form of a touch screen primary display and a button deck comprising a programmable LCD screen.
73. The gaming method of claim 72 wherein the programmable LCD screen is configured to depict a different arrangement of potential inputs corresponding to each of the two or more stored virtual casino table games.
74. The gaming method of claim 65 further comprising a selectable chip denomination module whereby an operator may select denominations of virtual betting chips depicted on the primary display.
75. The gaming method of claim 65 wherein only virtual casino table games are stored.
76. The gaming method of claim 65 wherein each game state engine includes a math model associated with the corresponding stored virtual table game.
77. The gaming method of claim 65 wherein an upper display area and a lower display area are located on two or more displays.
78. The gaming method of claim 77 wherein an upper display area and a lower display area are located on two or more displays with the displays positioned in a landscape orientation.
79. The gaming method of claim 65 wherein an upper display area and a lower display area are located on a single display.
80. The gaming method of claim 79 wherein the single display is positioned in a portrait orientation.
81. A gaming method comprising: utilizing a monetary input device configured to receive a physical item associated with a monetary value; enabling a player to select a wager for a game of chance and enabling the player to initiate a cash out operation; configuring at least one processor running executable instructions to operate a game of chance on a gaming machine including one or more displays and memory in communication with the at least one processor; utilizing one or more displays with at least a primary display area for playing virtual casino table games; configuring one or more user interfaces configured, along with the processor and associated memory, to allow a player to at least select a virtual casino table game to play from two or more stored virtual table games, place bets and start a hand or round of the selected virtual casino table game; configuring the processor running executable instructions to utilize a plurality of game state engines corresponding to at least two or more stored virtual casino table games including at least two table games from a group of table games consisting of craps, baccarat, roulette, blackjack, three-card poker, pai gow poker, Caribbean stud and big 6 wheel, wherein a sequence of play for each virtual casino table game is controlled by a dedicated one of the plurality of game state engines; utilizing an upper display area to depict at least a portion of game play while a lower display area is used to depict at least a portion of a digital betting game layout for at least one of the table games; and offering at least one side bet to the player with at least one side bet having a higher hold percentage than the associated virtual casino table game.
82. The gaming method of claim 81 wherein at least one of the at least one side bets increases the average hold percentage of the associated virtual casino table game when played.
83. The gaming method of claim 81 wherein at least one of the at least one side bets has a payout

which is a multiple of the associated virtual casino table game side bet wager.

84. The gaming method of claim 81 wherein at least one of the at least one side bets has a payout which is based on a progressive value of the associated virtual casino table game side bet.

85. The gaming method of claim 84 wherein the seed amount of the progressive value is greater than the side bet wager amount.

86. The gaming method of claim 81 wherein the plurality of side bets increase the average overall hold percentage of the gaming system when played.

87. The gaming method of claim 86 wherein the increase of the average overall hold percentage is based at least in part on a progressive virtual casino table game side bet when played.

88. The gaming method of claim 81 further comprising two user interfaces in the form of a touch screen primary display and a button deck comprising a programmable LCD screen.

89. The gaming method of claim 88 wherein the programmable LCD screen is configured to depict a different arrangement of potential inputs corresponding to each of the two or more stored virtual casino table games.

90. The gaming method of claim 81 further comprising a selectable chip denomination module whereby an operator may select denominations of virtual betting chips depicted on the primary display.

91. The gaming method of claim 81 wherein only virtual casino table games are stored.

92. The gaming method of claim 81 wherein each game state engine includes a math model associated with the corresponding stored virtual table game.

93. The gaming method of claim 81 wherein an upper display area and a lower display area are located on two or more displays.

94. The gaming method of claim 93 wherein an upper display area and a lower display area are located on two or more displays with the displays positioned in a landscape orientation.

95. The gaming method of claim 81 wherein an upper display area and a lower display area are located on a single display.

96. The gaming method of claim 95 wherein the single display is positioned in a portrait orientation.